

An aphid-resistant plant metabolite as a candidate aphicide: Insight into the bioactivity and action mode of betulin against aphids

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eLife Assessment

This **important** study identifies a plant-derived metabolite, betulin, as an effective natural insecticide against aphids and uncovers its specific molecular target. The evidence is **compelling**, combining greenhouse and field efficacy trials with rigorous molecular, genetic, and electrophysiological approaches that converge on a conserved binding site in the aphid GABA receptor. While additional work is needed to fully assess potential off-target effects and ecological safety, the study provides a strong mechanistic foundation. These findings will be of interest to researchers in plant biology, chemical ecology, and sustainable pest management.

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Abstract

Pest-resistant plants usually utilize secondary metabolites to cope with insect infestation. Betulin, a key bioactive compound in aphid-resistant wild peach, possesses promising applications in crop protection. Here, betulin, in both greenhouse and field experiments, displayed excellent control efficacy against *Myzus persicae*. RNA-seq, qRT-PCR, and western blotting revealed that betulin significantly inhibited the expression of *MpGABR* (encoding a GABA_A receptor). Besides, RNAi-mediated silencing of *MpGABR* markedly increased aphid sensitivity to betulin. Furthermore, MST (microscale thermophoresis) and voltage-clamp assays indicated that betulin bound to *MpGABR* ($K_d = 2.24 \mu\text{M}$) and acted as an inhibitor of *MpGABR*. Molecular docking, mutagenesis and genome editing suggested that THR228 is a critical and highly conserved site in *MpGABR* that betulin binds to specifically, causing aphid death. Overall, the activity of betulin depends on specific targeting and inhibition of

MpGABR. Elucidating the mechanism of action of this peach-derived insecticide may offer a sustainable green strategy for aphid control.

1. Introduction

Over more than 400 million years of coevolution with herbivores, plants have developed a variety of morphological (Soujanya et al. 2023 [\[1\]](#)), biochemical (Zhou et al. 2024b [\[2\]](#)), and molecular (Wang et al. 2024 [\[3\]](#)) defense strategies against insect herbivore attack. Plant secondary metabolites are highly valuable natural compounds, offering abundant and potent biochemical defense driven by plant-insect coevolution (Jahan et al. 2025a [\[4\]](#)). Interestingly, wild relatives of crops with insect resistance usually accumulate large amounts of biologically active compounds (Alonso-Salces et al. 2022 [\[5\]](#); Wang et al. 2022a [\[6\]](#)). Accordingly, investigating the bioactivities and potential mechanisms of action of compounds derived from wild resistant germplasms may provide effective and sustainable strategies for pest control in crop protection.

Myzus persicae (Sülzer) is one of the most destructive sap-feeding pests worldwide. It is able to settle on more than 400 plant species globally, displaying extraordinary polyphagia and a wide range of hazards (Bass et al. 2014 [\[7\]](#)). In addition to ingesting phloem saps and secreting honeydew, *M. persicae* can transmit various plant viruses (Guo et al. 2022 [\[8\]](#)), resulting in severe losses in crop yields. To date, the principal control strategy for *M. persicae* has relied on synthetic insecticides, including pyrethroids, organophosphorus compounds, carbamates and neonicotinoids (Jiménez-Jiménez et al. 2019 [\[9\]](#); Martins et al. 2021 [\[10\]](#); Zhang et al. 2023 [\[11\]](#)). However, long-term irrational application of these chemical insecticides has led to environmental pollution and promoted the resistance of aphids to insecticides (Trocza et al. 2021 [\[12\]](#); Zhang et al. 2023 [\[13\]](#); Stará et al. 2024 [\[14\]](#)). Therefore, it is necessary to develop novel green insecticides as potential alternatives to synthetic insecticides. Fortunately, plant secondary metabolites are a “treasure trove of active compounds” that serve as a rich resource for the development of promising green insecticides.

Our previous study revealed that *Prunus davidiana*, a close wild relative of cultivated peach, displays strong resistance to *M. persicae* through the accumulation of high levels of betulin (Wang et al. 2022a [\[15\]](#); Wang et al. 2024 [\[16\]](#)). In addition, betulin, a lupane-type triterpene, possesses strong insecticidal activity and is a promising substance for the development of novel aphid-control insecticides. A number of studies have reported that betulin and its derivatives exhibit a wide range of pharmacological activities (Amiri et al. 2020 [\[17\]](#)). An immune stimulant, Ir-Bet, was prepared using iridium complex and betulin, which evoked ferritinophagy-enhanced ferroptosis, thereby activating anti-tumor immunity (Lv 2023 [\[18\]](#)). The anti-inflammatory effect of betulin has been reported in macrophages at lymphoma site in mice (Szlasa et al. 2023 [\[19\]](#)). Betulin has been found to improve hyperlipidemia and insulin resistance and decrease atherosclerotic plaques by inhibiting the maturation of sterol regulatory element-binding protein (Tang et al. 2011 [\[20\]](#)). Additionally, betulin and its derivatives have been found to exhibit insecticidal activity against *Plutella xylostella* L. (Huang et al. 2025 [\[21\]](#)), *Aedes aegypti* (de Almeida Teles et al. 2024 [\[22\]](#)), and *Drosophila melanogaster* (Lee and Min 2024 [\[23\]](#)). However, the insecticidal mechanism of betulin against aphids remains unclear.

Gamma-aminobutyric acid (GABA) receptors have been confirmed to be targets of terpenoids that impair neuronal function in insect herbivores (Guo et al. 2023 [\[24\]](#)). There are two types of GABA receptors: ionotropic (GABA_A) and metabotropic (GABA_B) receptors. The first GABA receptor subunit identified in insects is encoded by *Rdl*, which confers resistance to dieldrin in *Drosophila melanogaster* (Ffrench-Constant et al. 1991 [\[25\]](#)). Although a variety of terpenoids play roles as positive allosteric modulators or noncompetitive antagonists of GABA_A receptors (GABRs) (Guo et al. 2023 [\[26\]](#)), whether betulin targets the GABR in *M. persicae* needs to be further investigated. GABRs are heteropentameric ligand-gated ion channels in the central nervous system that conduct

chloride and bicarbonate ions and are the target of numerous drugs for the treatment of neuropsychiatric disorders (Thompson 2024 [↗](#)). GABRs are composed of 5 different types of subunits, each with 4 helical transmembrane domains (TM1-TM4), of which TM2 is located at the center of the pentamer and forms an ion channel (Fan et al. 2024 [↗](#)). Previous study has proposed that vertebrate and human GABR genes maintain a broad and conservative gene clustering pattern, while in invertebrates, this pattern is missing, indicating that these gene clusters were formed early in vertebrate evolution and were established after invertebrates diverged. Notably, invertebrates each possess a unique GABR gene pair, which are homologous with human GABR α and β subunits, suggesting that the existing GABR gene cluster evolved from an ancestral α - β subunit gene pair (Tsang et al. 2006 [↗](#)). During the coevolution of plants and insects, the duplications and amino acid substitutions in GABR may have been beneficial for adaptation to insecticides and terpenoid compounds (Guo et al. 2023 [↗](#)).

In this study, betulin was confirmed to have excellent control efficacy against *M. persicae* in both greenhouse and field experiments. RNA-seq, qRT-PCR, and western blotting assays revealed that betulin significantly inhibited the expression of *MpGABR* in aphids. In addition, RNAi-mediated silencing of *MpGABR* markedly increased aphid sensitivity to betulin. Furthermore, microscale thermophoresis (MST) and voltage-clamp assays indicated that betulin can bind to *MpGABR* ($K_d = 2.24 \mu\text{M}$) and act as an inhibitor ($\text{EC}_{50} = 20.66 \mu\text{M}$) of *MpGABR*. Molecular docking analysis suggested that betulin bound to *MpGABR* via the THR228 amino acid residue. This site is highly conserved across 11 species in the Aphididae family of Hemiptera and may be a critical specific binding site for betulin in *MpGABR* in aphids. Mutagenesis and genome editing experiments revealed that betulin bound specifically to this amino acid residue in aphids but not in *Drosophila*, resulting in aphid death. This study elucidated the insecticidal mechanism of betulin involving the targeting of *MpGABR*, providing a sustainable green strategy for aphid control. However, given that betulin may affect a wider range of organisms, it should be used with caution.

2. Materials and methods

2.1 Test insects

M. persicae populations were collected from tobacco (*Nicotiana tabacum* L.) leaves. Wild-type (WT) flies (*D. melanogaster*) were purchased from Fungene Biotechnology Co., Ltd., Jiangsu, China. The test insects were cultured in a greenhouse (25 °C, 70% relative humidity and 14 h light/10 h dark cycle) in Beibei, Chongqing, China.

2.2 Leaf bioassay

The toxicity of betulin against *M. persicae* was measured using a slip-dip bioassay as previously described (Wang et al. 2022a [↗](#)). Thirty apterous adult *M. persicae* individuals were gently transferred on tobacco (*N. tabacum*) leaves using a fine brush. The tobacco leaves were subsequently immersed in 0.1% (v/v) Tween-80 and 3% (v/v) acetone solution containing betulin or pymetrozine (as a positive control) at 7 doses (0, 0.0625, 0.125, 0.25, 0.5, 1, and 2 mg·mL⁻¹) for 5 seconds. After drying, the leaves with *M. persicae* were placed on wet filter paper and cultured on a water-soaked sponge for 48 h. Three replicates were tested for each treatment. Furthermore, the LC₃₀, LC₅₀ and LC₇₀ values of betulin were calculated using log-probit analysis via IBM SPSS Statistics (v.22.0, Chicago, IL, USA) (Zhou et al. 2024b [↗](#)).

2.3 Greenhouse and field bioassays

The control effects of betulin against *M. persicae* in the greenhouse and field were evaluated by a spraying-based method as described previously (Zhou et al. 2023 [↗](#)). The commercial insecticide pymetrozine and an aqueous solution containing 0.1% (v/v) Tween-80 and 3% (v/v) acetone were employed as positive and negative controls, respectively. Sprays of 0.1641 mg·mL⁻¹ (LC₅₀) betulin

and $1.0612 \text{ mg} \cdot \text{mL}^{-1}$ (LC_{50}) pymetrozine (Table S1) were prepared in aqueous solutions including 3% (v/v) acetone and 0.1% (v/v) Tween-80, respectively. Before the control effect assays, each tobacco (*N. tabacum* L.) seedling was pre-inoculated with 60 apterous adults of *M. persicae*. In the greenhouse trial, each treatment included 8 replications, and each replication consisted of 3 tobacco plants growing under controlled conditions as described above. The field trial (Beibei District, Chongqing, China.) was conducted under natural conditions with an average temperature of 20 °C. Each treatment had 8 replications, covering an area of 20 m², with 3 protected rows per replication. Furthermore, each tobacco plant was sprayed with 20 mL of the corresponding solution with an electric sprayer. Mortality was recorded at 1, 5, 9, and 14 days after treatment. The control efficacy of betulin against *M. persicae* was calculated according to the methods of Zhou (Zhou et al. 2023 [DOI](#)).

2.4 RNA-seq assay

After exposure to $0.1641 \text{ mg} \cdot \text{mL}^{-1}$ (48 h LC_{50}) betulin for 48 h, 100 apterous *M. persicae* adults were sampled for each of 3 replicates, and total RNA was extracted via TRIzol reagent (Tiangen, Beijing, China) to construct mRNA sequencing libraries. RNA-seq analysis was performed on the Ion Proton BGISEQ-500 platform (Beijing Genomic Institute, BGI, China). After the raw reads were filtered, the clean reads were aligned to the *M. persicae* reference genome, which was downloaded from the National Center for Biotechnology Information database, using HISAT2 (v2.1.0) (Kim et al. 2015 [DOI](#)). The expression levels of genes in each sample were normalized to the fragments per kilobase million mapped reads (FPKM) value. The differentially expressed genes (DEGs) between groups were identified using the fold change factor ($|\text{Log}_2(\text{FC})| > 1$). Functional enrichment analysis of DEGs was performed using the Gene Ontology (GO) and Kyoto Encyclopedia of Genes and Genomes (KEGG) databases.

2.5 Validation by quantitative real-time PCR (qRT-PCR)

The gene-specific primers used for qRT-PCR, which were designed with Primer Premier 5.0, are listed in Table S2. According to the manufacturer's specifications for the Bio-Rad iQTM SYBR[®] Green Supermix Kit, qRT-PCR was performed with the following conditions: 60 °C for 15 s, followed by 48 cycles of 60 °C for 30 s, and 72 °C for 30 s. The *ribosomal protein S18* gene (*RPS18*) was used as the reference gene to normalize the expression levels. The relative expression levels of the target genes were calculated using the $2^{-\Delta\Delta C_t}$ approach (Zhou et al. 2024b [DOI](#)). Three biological replicates were used for each assay.

2.6 Cloning, bioinformatics and phylogenetic analysis

The cDNA of *M. persicae* was synthesized from the extracted total RNA using the PrimeScript[®] 1st Strand cDNA Synthesis Kit (TaKaRa, Japan). The coding sequences of *MpGABR* (encoding GABA_A receptor), *MpGABRAP* (encoding GABA_A receptor-associated protein) and *MpGABRB* (encoding GABA_A receptor β subunit) were cloned from *M. persicae* cDNA using the corresponding specific primers (Table S2) and then inserted separately into the pGEM-T Easy vector. The recombinant plasmids were subsequently transferred into *Escherichia coli* DH5α competent cells for sequencing. BioXM (v2.7.1) was used to analyze the sequence information of these three genes, including the length, molecular weight, and isoelectric point of the deduced amino acid sequence. MEGA7 was used for multiple amino acid sequence alignment (ClustalW) and phylogenetic analysis (neighbor-joining method with 1000 replicates) of *MpGABR*.

2.7 RNAi assay

To knock down the target genes (*MpGABR*, *MpGABRAP*, and *MpGABRB*), *MpGABR*-, *MpGABRAP*-, and *MpGABRB*-dsRNA were artificially synthesized using the dsRNA Synthesis Kit (Thermo Scientific, Vilnius, Lithuania, EU). In accordance with a previously described method (Zhou et al. 2024a [DOI](#)), fresh tobacco leaves were cut into 3.0 cm-diameter discs and placed in an oven at 50 °C for 3 min. Subsequently, the dried leaf discs were exposed to 10 μL of diethyl pyrocarbonate-

treated nuclease-free water (DEPC-water), dsGFP (green fluorescent protein, 1000 ng·μL⁻¹, negative control), and *MpGABR*-, *MpGABRAP*-, or *MpGABRB*-dsRNA (1000 ng/μL⁻¹) at 25 °C for 5 h. After the leaf discs had absorbed the solution, they were placed on wet filter paper on a water-soaked sponge. Subsequently, 30 apterous adult *M. persicae*, starved for 24 h, were fed on the treated leaf discs for 48 h. The surviving aphids were collected after exposure for qRT-PCR. Three biological replicates were included for each treatment.

2.8 Prokaryotic expression of *MpGABR* and western blotting

To produce the *MpGABR* protein *in vitro*, an *Escherichia coli* expression system was constructed (Zhou et al. 2021 [\[1\]](#)). Briefly, the coding sequence of *MpGABR* was inserted into the expression vector PET-B2M, obtained from BGI (Shenzhen, China), and transformed into BL21 (*DE3*) competent cells. To induce the expression of the *MpGABR* protein, 2 mM isopropyl β-D-1-thiogalactopyranoside (IPTG) was used, and then, the transformed BL21 cells were incubated at 180 rpm and 30 °C for 24 h. Finally, the protein was extracted and purified using Solarbio GST-tag Purification Resin for subsequent assays.

The purified *MpGABR* protein resolved on a 15% SDS-PAGE (sodium dodecyl sulfate-polyacrylamide gel electrophoresis) gel and then transferred to a polyvinylidene fluoride (PVDF) membrane at 20 V for 1 h. Moreover, 1 × TBST (Tris-Borate-Sodium Tween-20) containing 5% fat-free milk powder was used to block the PVDF membrane. Then, the PVDF membrane was incubated with an antibody against *MpGABR* (polyclonal rabbit antibody, provided by GeneCreate Biotechnology) at 25 °C for 1 h. Subsequently, the membrane was further incubated with the corresponding horseradish peroxidase (HRP)-conjugated goat anti-rabbit immunoglobulin G (IgG) antibody at 25 °C for 1 h and then washed three times with TBST. Finally, the bands on the membrane were visualized using a Chemi Doc MP Imaging System (Bio-Rad) after incubation with a chemiluminescence reagent (ECL, Bio-Rad). GAPDH was used as a reference protein.

2.9 Molecular docking

In accordance with a previously reported method (Zhou et al. 2023 [\[2\]](#)), the protein structure and function of *MpGABR* were predicted via the I-TASSER server (Iterative Threading ASSEMBLY Refinement). AutoDockTools software (v1.5.7) was used to conduct molecular docking. Furthermore, PyMOL (v3.1) and Discovery Studio (v4.5) were used to evaluate the molecular docking model.

2.10 Site-directed mutagenesis

The coding sequence of *MpGABR* was inserted into the PET-B2M vector and used as the template for site-directed mutagenesis. In accordance with the manufacturer's specifications, the Fast MultiSite Mutagenesis Kit (TransGen, Beijing, China) was used to generate the multisite mutations in *MpGABR* (Ma et al. 2020 [\[3\]](#)). Then, the *MpGABR* mutants, with the mutations of R224A (ARG224 replaced by ALA), A226T (ALA226 replaced by THR), F227Y (PHE227 replaced by TYR), or T228R (THR228 replaced by ARG), were amplified, and the amplicons were used to replace the *Bam*HI-*Sma*I fragment in the recombinant *MpGABR* plasmid.

2.11 Method for *Drosophila* genome editing

To generate *Drosophila* mutants bearing the *DmGABR*^{R122T} mutation, homologous to THR228 in *MpGABR*, the Clustered Regularly Interspaced Short Palindromic Repeats-Cas9 (CRISPR/Cas9) genomic editing strategy was used. First, CRISPR Optimal Target Finder (<http://targetfinder.flycrispr.neuro.brown.edu/> [\[4\]](#)) was used to generate short guide RNA (sgRNA) targets without off-target hits. Then, the Precision gRNA Synthesis Kit (Thermo Scientific) was used to synthesize the sgRNA. The 228 bp homologous arm in R122T, for homology-directed repair, was subsequently employed to synthesize single-stranded DNA (ssDNA) bearing the desired mutation (*DmGABR*^{R122T}). Subsequently, 1 nL of a mixture containing 160 ng·μL⁻¹ ssDNA, 300 ng·μL⁻¹ both sgRNAs, and 90

ng· μL^{-1} Cas9 protein was injected into *Drosophila* embryos via the Drummond Nanoject III (Broomall, PA, USA) system. CRISPR/Cas9-induced mutations were verified by amplifying and sequencing genomic DNA flanking the genome editing target site. Finally, according to the traditional topical application and bioassay approach described previously (Zhou et al. 2023 [\[1\]](#)), the acute insect toxicity of betulin in genome-modified or nonmodified (WT) 3-day-old adult flies was determined. The oligonucleotide sequences used for PCR, the sgRNA synthesis template and the fragment of donor DNA used for homology-directed repair (HDR) are listed in Table S3.

2.12 Microscale thermophoresis (MST) assay

The binding affinity of MpGABR for betulin was measured via a NanoTemper Technologies Monolith Kit (MO, Munich, Germany) according to the manufacturer's specifications. Briefly, MpGABR was fluorescently labeled via a protein labeling kit (RED-NHS 2nd generation, Monolith™) and then mixed with sixteen concentrations (from 5 nM to 60 mM) of betulin. The mixtures were subsequently loaded into NanoTemper capillaries for MST measurements with 40% LED and 60% MST power at 25 °C. MO Control NanoTemper Technologies GmbH software (v2.3) and MO Affinity Analysis software (v2.3, NanoTemper Technologies GmbH) were used to acquire and analyze the recorded data, respectively. Three replicates were set for each concentration. The dissociation constant (K_d), indicating the binding affinity, was calculated using data from three replicates for each experiment.

2.13 Electrophysiological recording

In accordance with previously described methods (Chen et al. 2019 [\[2\]](#)), oocytes isolated from mature healthy *Xenopus* females were digested at 25 °C for 1 h to remove the follicular layer. Individual oocytes were selected to express MpGABR and incubated at 18 °C with 1× Ringer's solution (96 mM NaCl, 5 mM MgCl₂, 5 mM HEPES, 2 mM KCl, and 0.8 mM CaCl₂) supplemented with 550 mg·mL⁻¹ sodium pyruvate, 100 mg·mL⁻¹ streptomycin, 50 mg·mL⁻¹ tetracycline, and 5% dialyzed horse serum. After 3 days of incubation, the oocytes were used for electrophysiological recording. GABA was dissolved in 1 × Ringer's solution to prepare 100 μM GABA solution. Subsequently, the 100 μM GABA solutions containing different concentrations of betulin (0, 5, 10, 20, 40, 80, 160, 320 μM) were used to perfuse the oocytes. The whole-cell currents were recorded via a two-electrode voltage-clamp. The data were collected and analyzed using Digidata 1440A and pCLAMP 10.2 software (Axon Instruments Inc., Union City, CA, USA), respectively (Xu et al. 2020 [\[3\]](#)).

2.14 Statistical analysis

Statistical analysis was performed using SPSS (v22). Statistical significance was determined using one-way analysis of variance (ANOVA) for differences among groups. The replicate numbers are indicated by the black dots in the figures.

3. Results

3.1 Control efficacy of betulin against aphids

Bioassays revealed that the LC₅₀ values of betulin and pymetrozine against *M. persicae* at 48 h were 0.1641 and 1.0612 mg·mL⁻¹, respectively (Table S1). To assess the control efficacy of betulin, tobacco plants infected with *M. persicae* were exposed to betulin (0.1641 mg·mL⁻¹) and pymetrozine (1.0612 mg·mL⁻¹, as a positive control) in both the greenhouse and field. In the greenhouse test (Fig. 1A-D [\[4\]](#)), at 14 days after treatment, *M. persicae* in the CK group (negative control) reproduced rapidly, and tobacco leaves turned yellow and wrinkled, whereas tobacco leaves in the betulin and pymetrozine treatment groups were barely affected by the aphids (Fig. 1A-B [\[4\]](#)). The control efficacy of betulin against *M. persicae* was significantly greater than that of pymetrozine at 1 day after treatment ($P < 0.01$), while there were no significant differences

between the control efficacies of betulin and pymetrozine at 5, 9, or 14 days after treatment (**Fig. 1E**). After 14 days of treatment, the control efficacies of betulin and pymetrozine reached 91.54% and 95.15%, respectively. Throughout the field test (**Fig. 1F**), no difference in the control efficacy was observed between betulin and pymetrozine, with values of 87.79% and 90.14%, respectively, after 14 days of treatment (**Fig. 1G**). These results suggest that betulin has immense potential for development as a commercial aphid insecticide like pymetrozine.

3.2 Expression patterns of genes in aphids after treatment with betulin

To investigate the expression patterns of genes in *M. persicae* after exposure to betulin, RNA-seq was performed on *M. persicae* with 0.1641 mg·mL⁻¹ betulin treatment for 48 h and without (CK) (**Fig. 2A**). Principal coordinate analysis (PCA) revealed that the expressed genes were strongly clustered in the CK group and betulin group (**Fig. 2B**). Compared with those in the CK group, there were 130 up- and 41 downregulated significant differentially expressed genes (DEGs) in *M. persicae* after betulin treatment (**Fig. 2C and D**, Table S4). To validate the RNA-seq results, fifteen DEGs were randomly selected with different expression levels. The relative expression trends of the fifteen DEGs determined via qRT-PCR were similar to those detected via RNA-seq, supporting the reliability of the RNA-seq data (**Fig. 2E**). To further analyze the functions of the DEGs, GO (Gene Ontology) and KEGG (Kyoto Encyclopedia of Genes and Genomes) enrichment analyses were performed. GO term enrichment demonstrated that the DEGs were enriched mainly in the GABA_A receptor activity term (**Fig. 2F**). Interestingly, KEGG enrichment also found that the DEGs were enriched in the GABAergic signaling related pathway and GABAergic synapse terms (**Fig. 2G**). Additionally, the expression of DEGs related to GABR, including *MpGABR* (encoding GABA_A receptor), *MpGABRAP* (encoding GABA_A receptor-associated protein) and *MpGABRB* (encoding GABA_A receptor β subunit), in the betulin group was significantly lower than that in the CK group (**Fig. 2H**). In particular, the log₂(fold change) value of *MpGABR* was the lowest, at -3.8 (Table S4). These results suggested that *MpGABR* may be a candidate target for betulin against *M. persicae*.

3.3 *MpGABR* expression was inhibited by betulin

Analysis of the sequence information revealed that the *MpGABR*, *MpGABRAP* and *MpGABRB* proteins have 694, 118, and 250 amino acids, respectively, with calculated molecular weights of 77.20, 14.08, and 28.01 kDa and isoelectric points of 9.85, 9.63, and 7.20, respectively (Table S5). Additionally, structure prediction indicated that *MpGABR*, *MpGABRAP* and *MpGABRB* contain 4, 0, and 1 transmembrane helical domain, respectively (**Fig. 3A**), implying that only *MpGABR* has a complete transmembrane structure.

To further verify the inhibitory effect of betulin on GABR-related gene expression, the relative expression of the three GABR-related genes was detected in *M. persicae* exposed to the LC₃₀, LC₅₀, and LC₇₀ of betulin for 48 h. The relative expression of all three genes decreased gradually as the concentration of betulin increased (**Fig. 3B-D**). After exposure to the LC₃₀ of betulin for 48 h, the relative expression of *MpGABR*, *MpGABRAP* and *MpGABRB* decreased by 82.91%, 10.53%, and 11.49%, respectively. These results indicated that *MpGABR* was the most sensitive to betulin. To further investigate the effect of betulin on the *MpGABR* protein, *M. persicae* was exposed to betulin concentrations of LC₃₀, LC₅₀, and LC₇₀ for 48 h. As the concentration of betulin increased, the *MpGABR* protein content in *M. persicae* gradually decreased (**Fig. 3E and F**). Compared with that in the CK group, the *MpGABR* protein content in *M. persicae* exposed to betulin concentrations of LC₃₀, LC₅₀, and LC₇₀ decreased by 33.68%, 44.89%, and 76.89%, respectively.

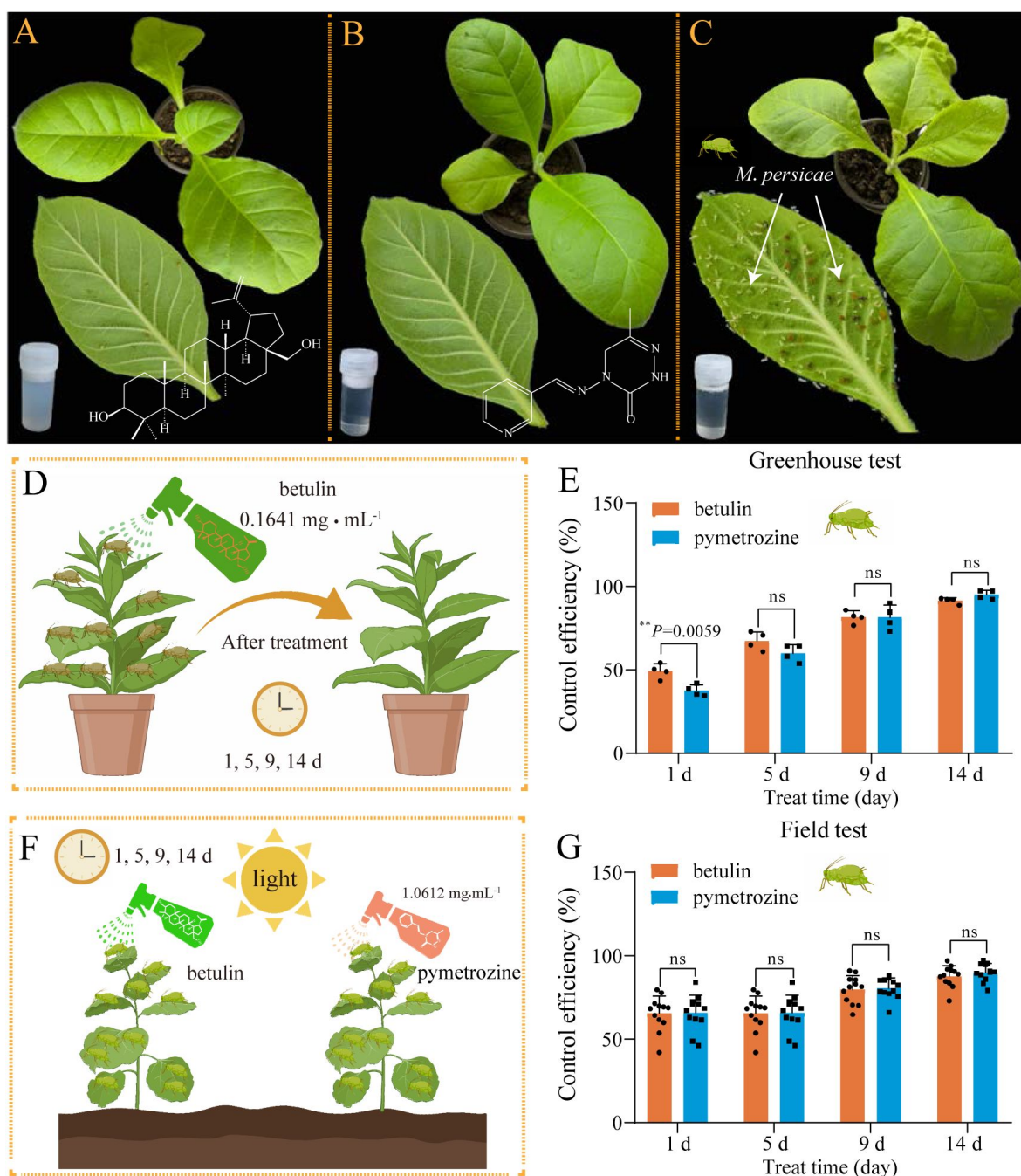


Figure 1.

The aphicidal activity of plant-derived derivative betulin against *M. persicae*.

(A-G) Representative images of the control effects of betulin (A, $0.1641 \text{ mg} \cdot \text{mL}^{-1}$, 48 h LC_{50} value of betulin), pymetrozine (B, $1.0612 \text{ mg} \cdot \text{mL}^{-1}$, a positive control), and CK (C, a negative control) against *M. persicae* in greenhouse (A-E) and field (F, G) tests after treatment for 14 d. The data are shown as the mean $\pm$ SD from twelve independent experiments. ** $P < 0.01$. ns, not significant.

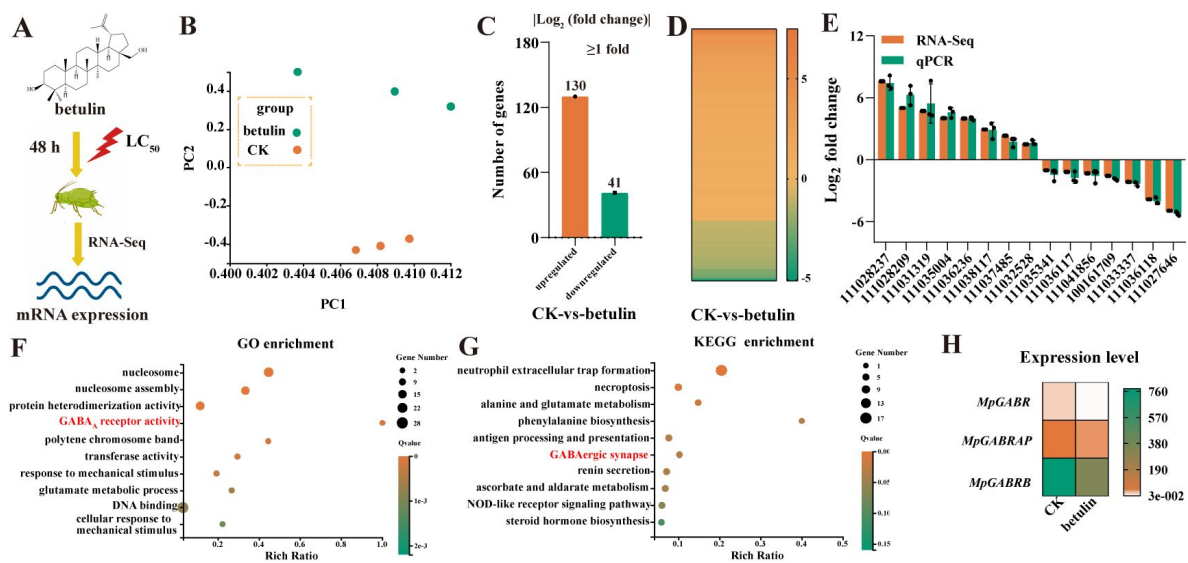


Figure 2.

RNA-Seq analysis revealed candidate targets for betulin against *M. persicae*.

(A) Diagram of the candidate targets for betulin against aphids as determined by RNA-Seq. Water containing 0.1% (v/v) Tween-80 and 3% (v/v) acetone was used as the control treatment (CK). (B) Principal coordinates analysis (PCA) analysis differentially expressed genes in RNA-Seq. (C-D) Distribution (C) and heatmap (D) of the significantly differentially expressed genes (DEGs) in *M. persicae*. (E) qPCR validation of 15 DEGs identified by RNA-Seq. (F-G) Top 10 enriched GO (F) and KEGG (G) pathways of the DEGs. The "rich ratio" was defined as the ratio of the number of DEGs enriched in the pathway to the total number of genes enriched in the same pathway. (H) Heatmap of the expression level of the GABR genes identified by RNA-Seq. *MpGABR*, encoding GABA_A receptor; *MpGABRAP*, encoding GABA_A receptor associated protein; *MpGABRB*, encoding GABA_A receptor β subunit.

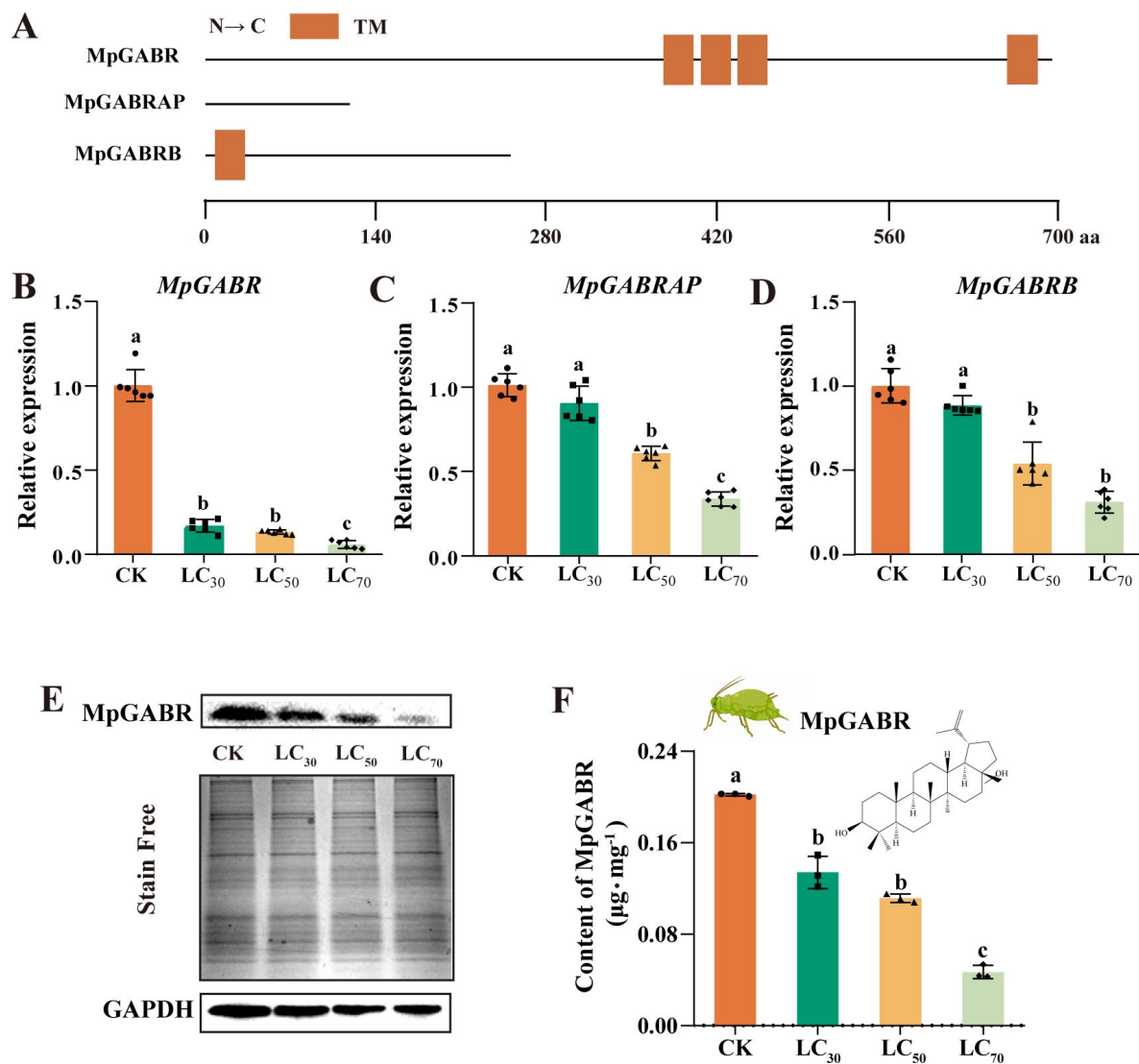


Figure 3.

***MpGABR* expression in aphids was significantly inhibited by treatment with betulin.**

(A) Schematic drawing of *MpGABR*, *MpGABRAP* and *MpGABRB*. TM, transmembrane helices. (B-D) qPCR expression analysis of *MpGABR* (B), *MpGABRAP* (C) and *MpGABRB* (D) transcripts in *M. persicae* exposed to LC₃₀, LC₅₀, and LC₇₀ of betulin for 48 h. The bars represent the average ($\pm$ SD). Different letters above the error bars indicate significant difference (ANOVA, Tukey's test, $P < 0.05$). (E, F) Western blotting analysis of *MpGABR* protein after betulin treatment for 48 h at three different concentrations (LC₃₀, LC₅₀ and LC₇₀). The bars represent the average ($\pm$ SD). Different letters above the error bars indicate significant difference (ANOVA, Tukey's test, $P < 0.05$).

Furthermore, after these three genes were silenced via RNAi (**Fig. 4D**), the expression levels of *MpGABR*, *MpGABRAP* and *MpGABRB* in *M. persicae* significantly decreased by 64.57%, 63.28%, and 67.33%, respectively, compared with those in the control groups (DEPC-treated water and dsGFP) ($P < 0.0001$, **Fig. 4A–C**). After exposure to the LC_{50} of betulin for 48 h, the mortality of *M. persicae* with *MpGABR* silenced markedly increased ($P < 0.001$) by 30.44%, compared with that in the control groups, whereas the mortalities of *M. persicae* in the *MpGABRAP*- and *MpGABRB*-silenced groups did not significantly differ from that in the control groups (**Fig. 4E**). Additionally, western blotting analysis revealed that *MpGABR* protein expression significantly decreased by 69.91% after RNAi ($P < 0.0001$, Fig. S1A, B). These results further implied that *MpGABR* might be a target for betulin against *M. persicae*.

3.4 Phylogenetic analysis of GABRs in insects

Furthermore, phylogenetic analysis of GABRs in insects was performed using the amino acid sequences of 70 GABR proteins from Hemiptera, Diptera, Lepidoptera, Thysanoptera, Hymenoptera, and Coleoptera. *MpGABR* (XP_022173711.1) was found to be genetically closely related to CAI6365831.1 from *Macrosiphum euphorbiae*, XP_008183008.2 from *Acyrtosiphon pisum*, and XP_060864885.1 from *Metopolophium dirhodum* (**Fig. 5** and Table S6), indicating that *MpGABR* may be functionally analogous to CAI6365831.1, XP_008183008.2, and XP_060864885.1.

3.5 Interaction between *MpGABR* and betulin

The WT (wild type) *MpGABR* was expressed in *Escherichia coli* (**Fig. 6A**). The binding affinity of betulin with WT *MpGABR* was further measured using MST. The K_d value was 2.24 μ M (**Fig. 6B**), suggesting that betulin could bind to *MpGABR*. Additionally, voltage-clamp-based electrophysiological recording demonstrated that *MpGABR* strongly responded to betulin (**Fig. 6C**). In the presence of only GABA, a fast inward current was generated. As the concentration of added betulin increased, the current and channel activity of *MpGABR* decreased correspondingly, with an EC_{50} value of 20.66 μ M (**Fig. 6C and D**). These results indicated that betulin acted as an inhibitor of *MpGABR*.

3.6 Key sites for the binding of betulin to *MpGABR*

To further study the binding mode of betulin in the active pocket of *MpGABR*, molecular docking was performed to explore the structure–function relationships between betulin and *MpGABR*. The docking results revealed that the predicted binding energy between betulin and *MpGABR* was $-6.38 \text{ kcal}\cdot\text{mol}^{-1}$ (Table S7), verifying that betulin could act as a specific ligand for *MpGABR*. The three-dimensional binding pattern of betulin with *MpGABR* indicated that the four key amino acid residues (ARG224, ALA226, PHE227 and THR228) interacted with betulin in the *MpGABR* binding pocket (**Fig. 7A**). Among the four key amino acid residues, ALA226 and THR228 interacted with betulin via hydrogen bonding in the *MpGABR* binding pocket (Table S7). Betulin generated nonconventional H-bonds (C...H) with ALA226 (3.31 Å) and conventional H-bonds (C–O H) with THR228 (2.16 Å). Furthermore, sequence alignment of the key amino acid residues of GABR in different species from Hemiptera, Diptera, Lepidoptera, Hymenoptera, Thysanoptera, and Coleoptera indicated that, among the four residues, only THR228 was conserved across 11 species in the Aphididae family of Hemiptera (**Fig. 7B**), implying that THR228 may be an essential specific site for the action of betulin on *MpGABR* in aphids. Mutants of aphid *MpGABR* were subsequently constructed, including R224A (ARG224 replaced by ALA), A226T (ALA226 replaced by THR), F227Y (PHE227 replaced by TYR, similar to the site in *Drosophila*), or T228R (THR228 replaced by ARG, similar to the site in *Drosophila*) (**Fig. 7C**). The binding affinity of betulin with these mutants was measured using MST. The results revealed that the K_d values of R224A, A226T, F227Y, T228R, and WT were 2.31, 2.27, 2.25, 5321, and 2.24 μ M, respectively (**Fig. 7D and E**, Table

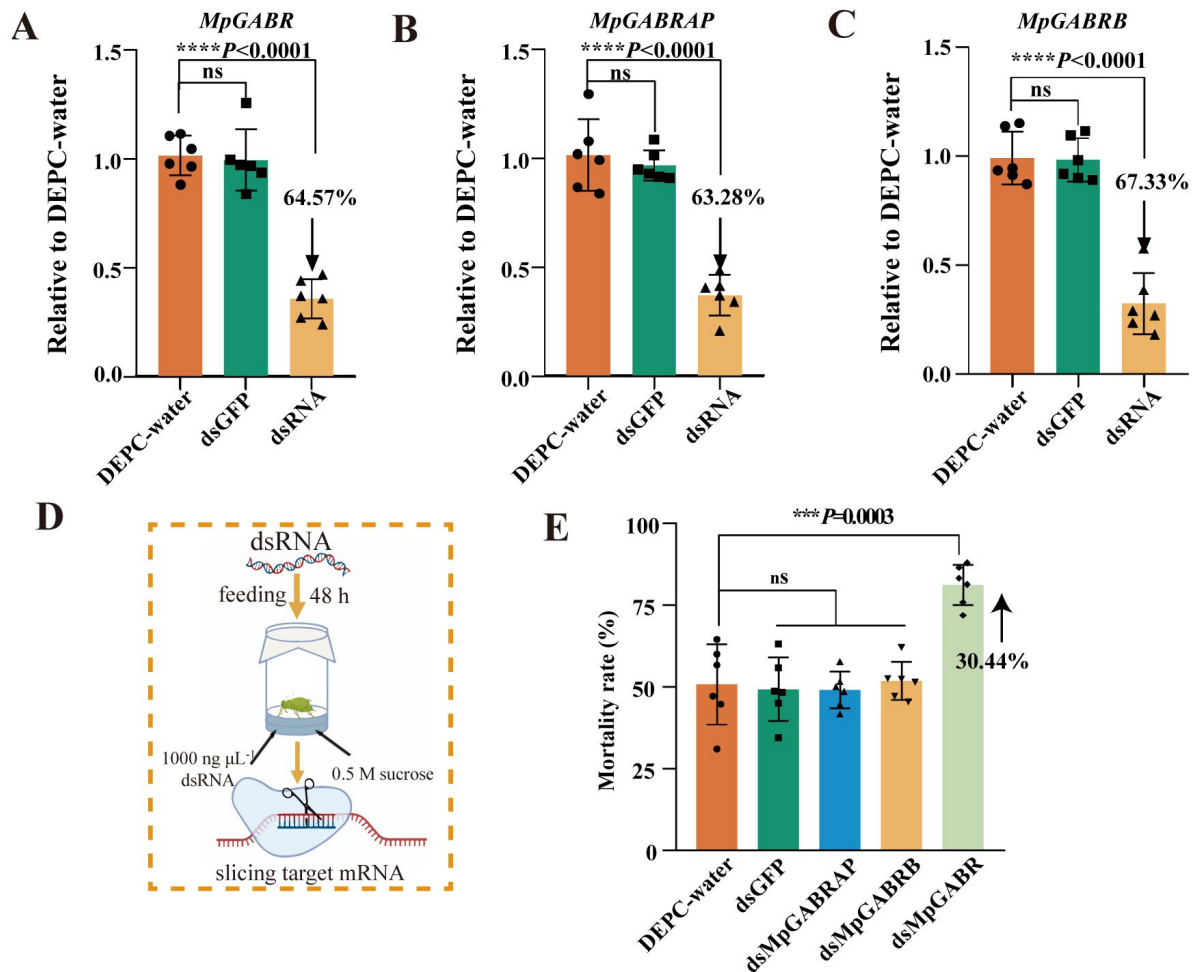


Figure 4.

Silencing the expression of *MpGABR*, *MpGABRAP* and *MpGABRB* via RNAi.

(A-C) qPCR expression analysis of *MpGABR* (A), *MpGABRAP* (B) and *MpGABRB* (C) after RNAi at 48 h post-dsRNA feeding relative to the expression levels after DEPC-water treatment. (D) Schematic drawing of the RNAi assay in *M. persicae*. (E) Mortality of aphids exposed to the LC_{50} of betulin for 48 h after RNAi. An asterisk (*) on the error bar indicates a significant difference between the treatment and group CK according to *t* tests, *** $P < 0.001$, **** $P < 0.0001$. ns, not significant.

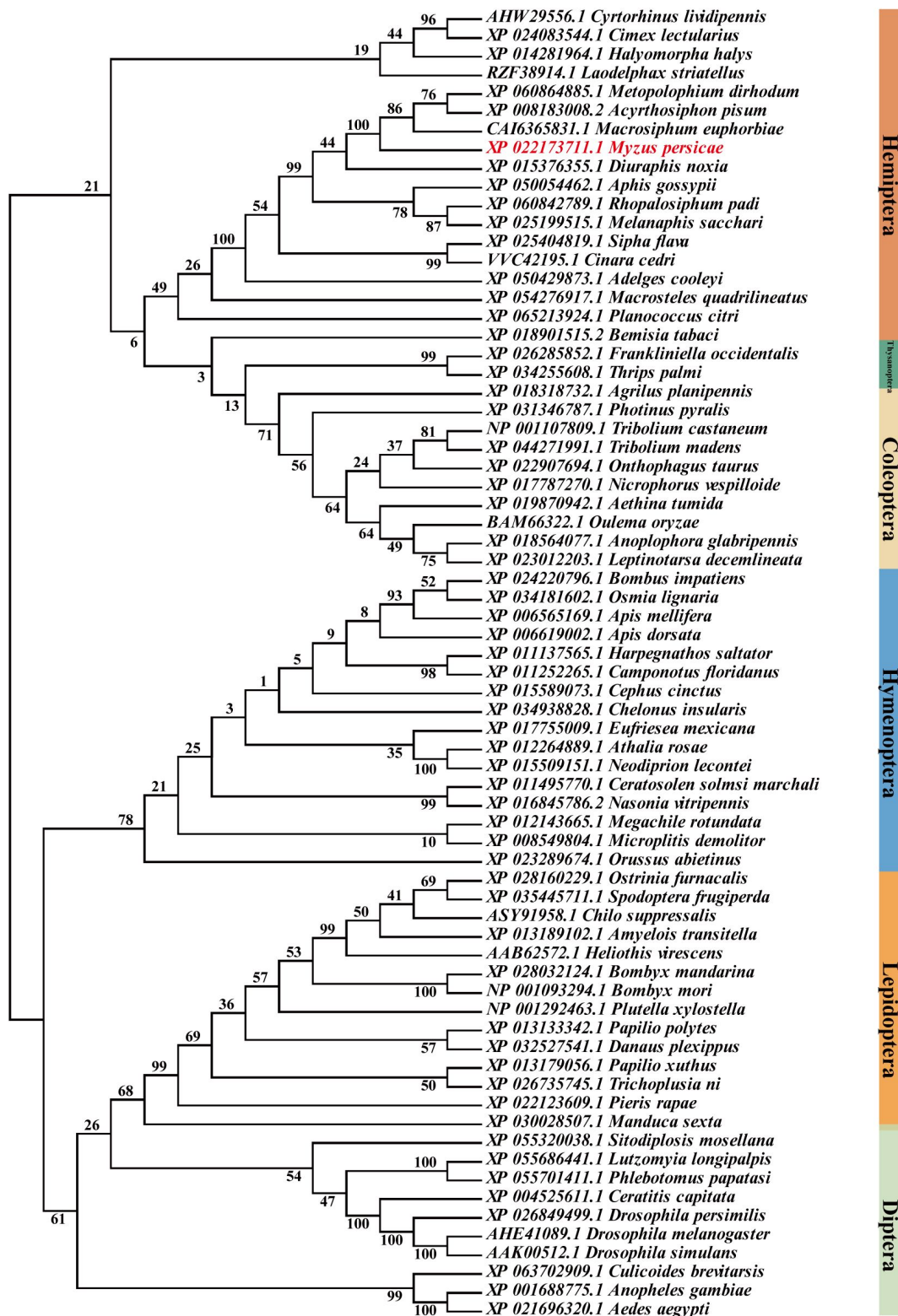


Figure 5.

Phylogenetic analysis of GABR in insects.

Neighbor-joining method with 1000 replicates based phylogenetic tree of GABR from six orders: Hemiptera, Diptera, Lepidoptera, Thysanoptera, Hymenoptera, and Coleoptera. Protein sequence alignment was performed using ClustalW in MEGA 7. The sequences used for constructing the tree were listed in Table S6.

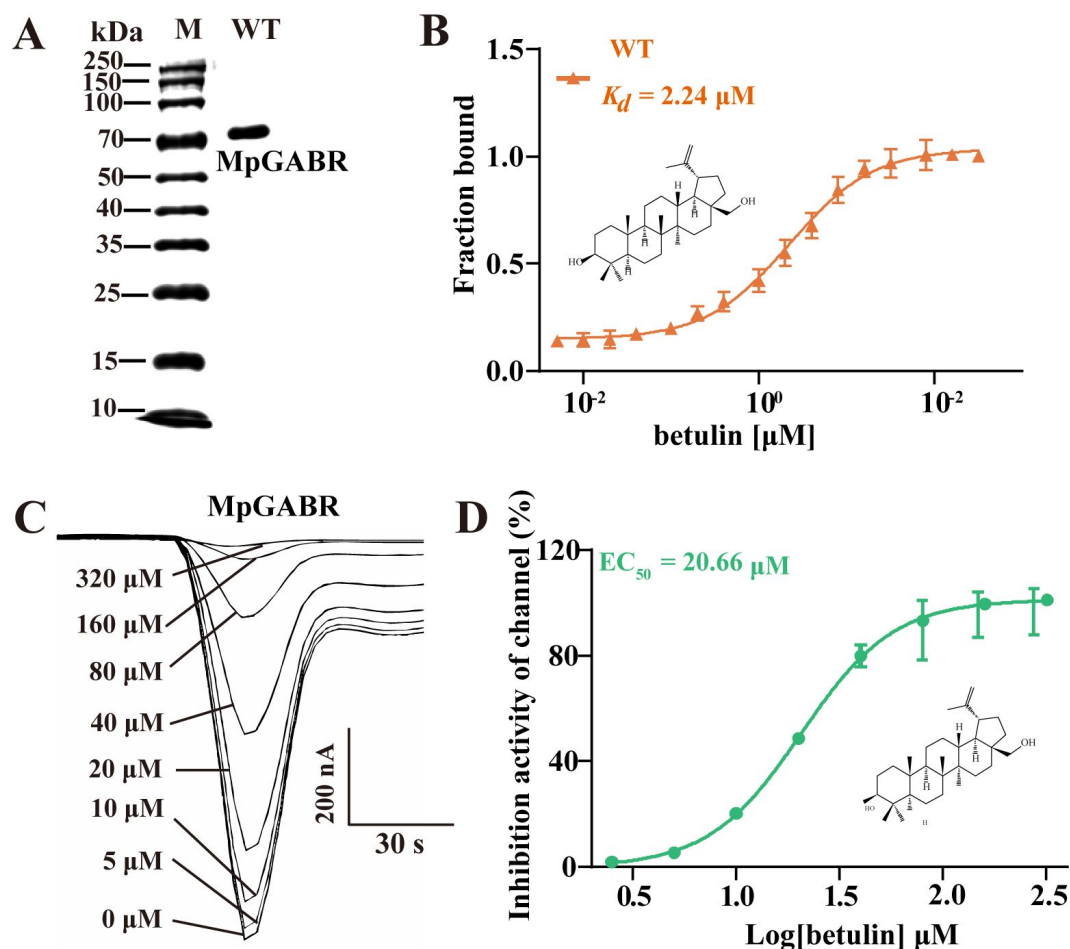


Figure 6.

The interaction of betulin with MpGABR protein.

(A) Expression of recombinant wild type-aphid MpGABR (1, WT) protein by *Escherichia coli*. (B) Quantification of the binding affinity of betulin with wild type-aphid MpGABR using MST. Current responses (C) and inhibition activity (D) of MpGABR induced by different concentration of betulin in the presence of GABA. 0 μM MpGABR indicates the presence of only GABA.

S8). This result suggested that the mutation at the THR228 site, with the highest K_d value, markedly reduced the binding ability of betulin to MpGABR, implying that the THR228 site may be an essential specific site for the binding of betulin to MpGABR.

3.7 Betulin binds specifically to MpGABR via THR228

To further prove that THR228 is the specific binding site for betulin in MpGABR, the WT and mutant (R122T, equivalent to THR228 in MpGABR) *Drosophila* DmGABR proteins were expressed in *E. coli* and extracted (Fig. 8A and D). The binding affinities of betulin and pymetrozine with DmGABR^{WT} and DmGABR^{R122T} were evaluated using MST. As shown in Fig. 8B, betulin was able to bind to DmGABR^{R122T} ($K_d = 342.4 \mu\text{M}$) but not to DmGABR^{WT}. However, pymetrozine exhibited no binding affinity for DmGABR^{R122T} or DmGABR^{WT} (Fig. 8C). Moreover, to generate *Drosophila* mutant individuals carrying DmGABR^{R122T}, the CRISPR/Cas9 genomic editing strategy was used. After exposure to different concentrations of betulin, the mortality rate of DmGABR^{R122T} *Drosophila* was significantly greater than that of DmGABR^{WT} *Drosophila* (Fig. 8E). In addition, the LD₅₀ value of betulin against DmGABR^{R122T} *Drosophila* at 72 h was 27.19 $\mu\text{g}\cdot\text{fly}^{-1}$, whereas the LD₅₀ value of betulin against DmGABR^{WT} *Drosophila* could not be calculated because the lethal concentration exceeded 1000 $\mu\text{g}\cdot\text{fly}^{-1}$ (Table S9). Additionally, there was no significant difference in mortality between DmGABR^{WT} and DmGABR^{R122T} *Drosophila* after exposure to pymetrozine (Fig. 8F). The LD₅₀ values of pymetrozine against DmGABR^{WT} and DmGABR^{R122T} *Drosophila* at 72 h were 0.23 and 0.24 $\mu\text{g}\cdot\text{fly}^{-1}$, respectively (Table S9). Moreover, THR228 is located at the neurotransmitter-gated ion-channel ligand-binding domain in MpGABR (Fig. S2), implying that betulin may play a role as a competitive antagonist of MpGABR. Taken together, these findings suggested that betulin bound specifically to MpGABR via THR228, resulting in the death of *M. persicae*.

4. Discussion

Although chemical pesticides are effective in pest control, the long-term unreasonable application of these substances has led the emergence of resistant pests, environmental deterioration, and deleterious effects on nontarget organisms, including humans, raising widespread and intense concerns (Devi et al. 2022). These challenges have made it urgent for the development of alternative strategies for pest control. Recently, the utilization of plant secondary metabolites as insecticides has become increasingly popular as an eco-friendly and biocontrol approach (Ling et al. 2022; Ayilara et al. 2023; Jahan et al. 2025b). Our previous study revealed that *P. davidiana*, a wild relative of cultivated peach, strongly resists *M. persicae* by accumulating high contents of betulin (Wang et al. 2022a; Wang et al. 2024). In addition, betulin, a lupane-type triterpene, possesses potent insecticidal activity and is a promising substance for the development of novel insecticides for aphid control. In this study, the insecticidal effect of betulin was further evaluated by comparing the control efficacy of betulin with that of pymetrozine against aphids in greenhouses and fields (Fig. 1A-G). These results indicate that betulin has a similar control effect to pymetrozine and has immense potential for development as a plant-derived insecticide.

Terpenes are a diverse group of plant secondary metabolites that can increase the resistance of plants to insect herbivores through direct (Wang et al. 2025b) and indirect (Wang et al. 2025a) defense mechanisms. In direct defense against herbivores, triterpenes play important roles in diverse biological activities, including antiparasitic, insecticidal, and antifeedant activities (Tian et al. 2021; Ivana et al. 2023). Azadirachtin, a tetracyclic triterpenoid compound isolated from the Indian neem tree (*Azadirachta indica*), is one of the most prominent commercial biopesticides, exhibiting strong insect antifeedant properties as well as growth- and reproduction-regulating effects (Dawkar et al. 2019; Bae et al. 2022). Besides, triterpene glycoside compounds play crucial roles in the defense of tobacco (*N. attenuata*) against tobacco hornworm (*Manduca sexta*)

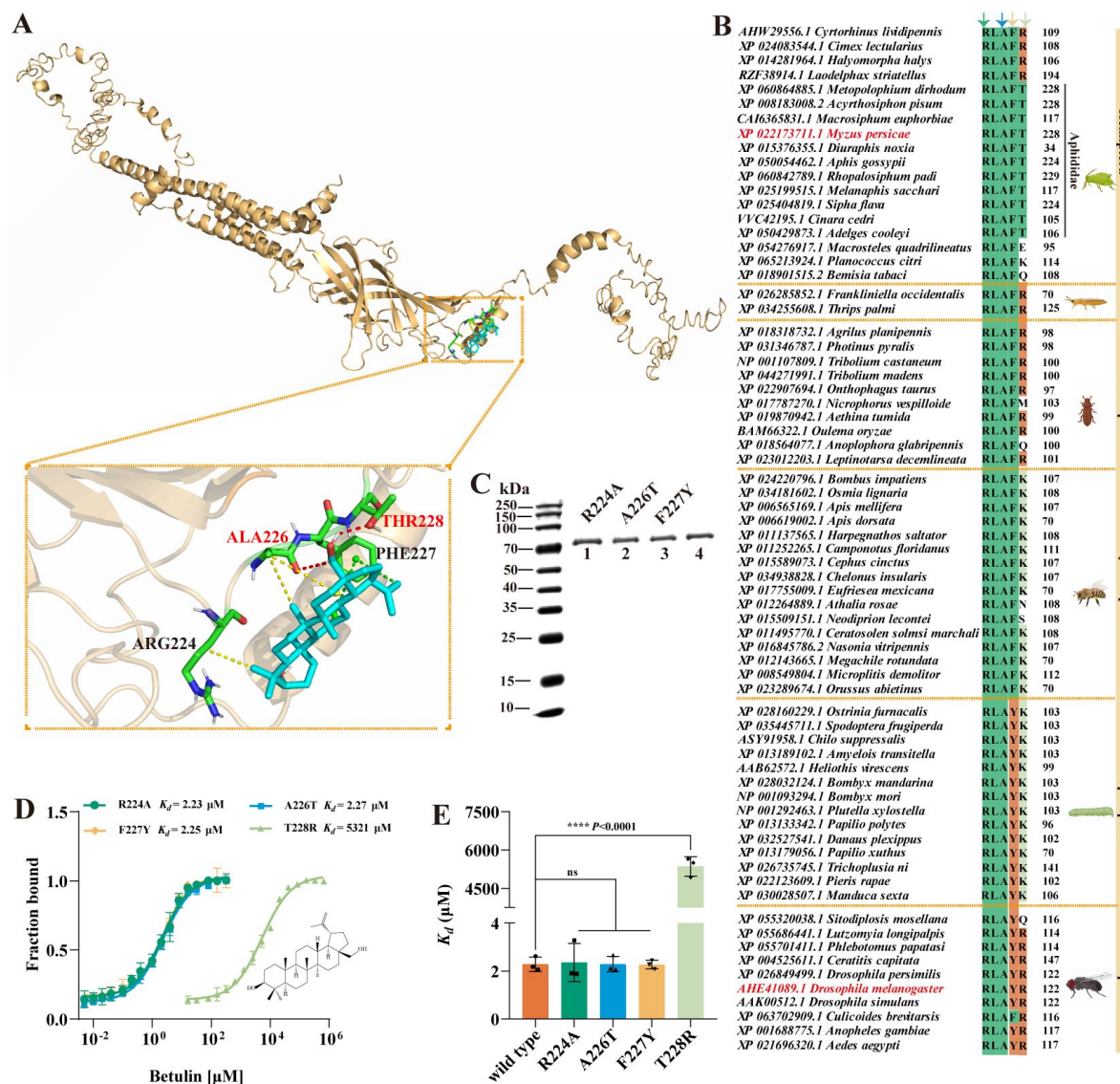


Figure 7.

Molecular docking, binding site and inhibitory effect of betulin on MpGABR.

(A) Best conformations of betulin docked to the binding pocket of MpGABR in aphids. An enlarged view of the betulin binding sites in MpGABR is indicated by a dashed frame. Potential Pi-alkyl (green), Alkyl (yellow), and hydrogen bond (red) interactions are indicated by dashed lines. (B) Sequence alignment of the key amino acids bound to betulin. The conserved residues among the different species in GABR are shown in orange, cyan, and light green. The numbers next to the amino acid indicates the site of the last residue of the key amino acids. (C) Expression of the recombinant mutation type-aphid MpGABR by *Escherichia coli*. Lane 1: R224A, Lane 2: A226T, Lane 3: F227Y, Lane 4: T228R. (D, E) Quantification of the binding affinity of betulin with wild-type and mutant type-aphid MpGABR using MST. The bars represent the average ($\pm$ SD) values. * indicates a significant difference (Student's *t* test, **** $p < 0.0001$).

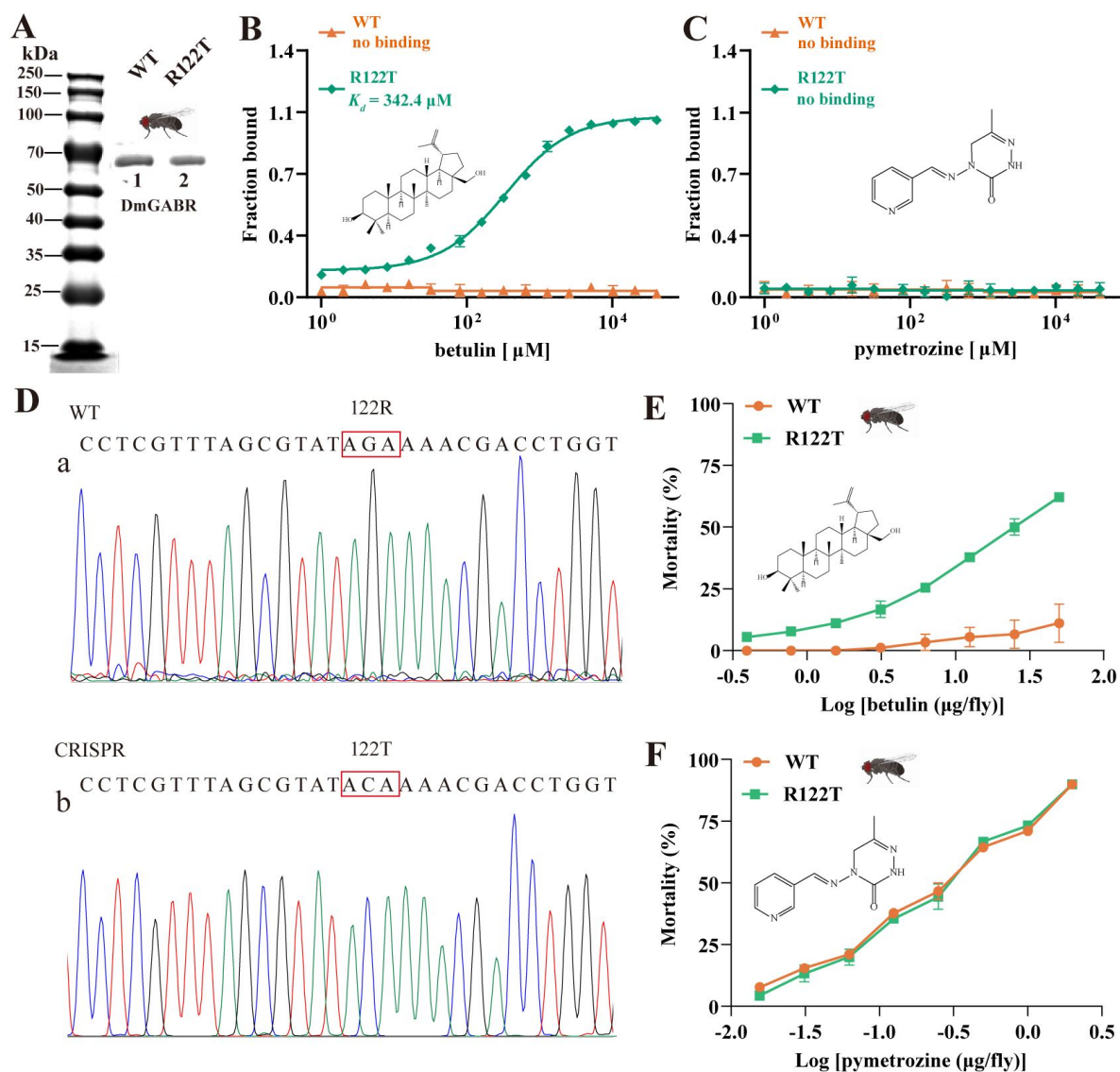


Figure 8.

Gene editing in *Drosophila* validated the species-specific binding site of betulin.

(A) Expression of the recombinant wild type (WT) and mutation type (R122T) of DmGABR (*Drosophila melanogaster*) by *E. coli*. Lane 1: wild type, Lane 2: R122T. (B, C) Quantification of the binding affinity of betulin (B) and pymetrozine (C) with WT and R122T of DmGABR using MST. (D) Sanger sequencing of the *DmGABR* gene in flies. Direct sequencing chromatograms of PCR products amplified from a fragment of gDNA flanking the WT (a), and introduced R122T (b) mutant flies. (E-F) Toxicity curves of betulin (E), and pymetrozine (F) against WT, and the DmGABR^{R122T} of the Cas9 *Drosophila*.

larvae (Yang et al. 2023 [↗](#)). Although betulin exhibits various pharmacological activities (Amiri et al. 2020 [↗](#); Li et al. 2022 [↗](#); Yan 2022 [↗](#); Lv 2023 [↗](#)), reports on its insecticidal activity are limited. Encouragingly, the effects of betulin and its derivatives on pests have attracted increasing attention. Betulinic acid and its derivatives showed larvicidal activity against *Aedes aegypti* larvae (da Silva et al. 2016 [↗](#)). Betulin-cinnamic acid-related hybrid compound 5b exhibited strong aphicidal activity, and compound 2l could destroy the ultrastructure of midgut cells and significantly inhibit the activity of α -amylase in diamondback moth (*Plutella xylostella* L.) larvae (Huang et al.). Our previous studies also indicated that betulin possesses potent insecticidal activity and is a key endogenous secondary metabolite related to the defense of peach against *M. persicae* (Wang et al. 2022a [↗](#)). Elucidating the insecticidal mechanism of betulin against aphids will provide a basis for the development of novel aphicides and sustainable strategies for aphid control.

GABA receptors have been confirmed to be targets of terpenoids that impair insect neuronal function in herbivores (Guo et al. 2023 [↗](#)). GABRs are heteropentameric ligand-gated ion channels in the central nervous system that conduct chloride and bicarbonate ions. These receptors are targets of numerous drugs for the treatment of neuropsychiatric disorders (Thompson 2024 [↗](#)). A variety of terpenoids act as positive allosteric modulators or noncompetitive antagonists of GABRs, such as diterpenoids (isopimaric acid and miltirone), sesquiterpenoids (picrotoxin, bilobalide and ginkgolides) and monoterpenoids (α -thujone and thymol) (Guo et al. 2023 [↗](#)). In this study, both GO and KEGG enrichment analyses revealed that the DEGs identified by RNA-seq were enriched in GABAergic signaling-related pathways (Fig. 2F and G [↗](#)). Additionally, the expression of the DEGs related to GABRs, particularly *MpGABR* (Table S4), in the betulin group was significantly lower than that in the CK group (Fig. 2H [↗](#)). Besides, the relative expression of *MpGABR*, *MpGABRAP* and *MpGABRB* decreased gradually after *M. persicae* was exposed to the LC₃₀, LC₅₀, and LC₇₀ of betulin for 48 h (Fig. 3B-D [↗](#)). Among them, *MpGABR* was the most sensitive to betulin, and its expression was reduced by 82.91% after exposure to the LC₃₀ of betulin for 48 h. Furthermore, compared with the control group, the *M. persicae* group with *MpGABR* silenced by RNAi presented a significant increase in mortality ($P < 0.001$), by 30.44%, after 48 h of exposure to the LC₅₀ of betulin (Fig. 4E [↗](#)). Collectively, these results suggest that betulin may have insecticidal effects on aphids by inhibiting *MpGABR* expression. The regulation of gene expression is sophisticated and delicate (Pope and Medzhitov 2018 [↗](#)). The regulatory network controlling *GABR* expression remains unclear. In adult rats, epileptic seizures have been reported to increase the levels of brain-derived neurotrophic factor (BDNF), which in turn prompted the transcription factors CREB and ICER to reduce the gene expression of the GABR $\alpha 1$ subunit (Lund et al. 2008 [↗](#)). In *Drosophila*, it has been demonstrated that WIDE AWAKE, which regulated the onset of sleep, interacted with the GABR and upregulated its expression level (Liu et al. 2014 [↗](#)). In *Drosophila* brain, circular RNA circ_sxc was found to inhibit the expression of miR-87-3p in the brain through sponge adsorption, thereby regulating the expression of neurotransmitter receptor ligand proteins, including GABR, and ensuring the normal function of synaptic signal transmission in brain neurons (Li et al. 2024 [↗](#)). However, it remains unclear how betulin reduces *MpGABR* expression, and further research is needed.

Additionally, betulin has been reported to be able to bind to GABRs (Manayi et al. 2016 [↗](#)). We further investigated the interaction of betulin with the *MpGABR* protein. The MST assay revealed that betulin was able to bind to *MpGABR* ($K_d = 2.24 \mu\text{M}$) (Fig. 6B [↗](#)), which is consistent with previous findings showing that betulin binds to GABA receptors in mouse brains *in vitro* (Muceniece et al. 2008 [↗](#)). Voltage-clamp-based electrophysiological recordings indicated that betulin acted as an inhibitor ($\text{EC}_{50} = 20.66 \mu\text{M}$) for *MpGABR* (Fig. 6C and D [↗](#)). Subsequent molecular docking analysis suggested that four key amino acid residues (ARG224, ALA226, PHE227 and THR228) interact with betulin in the *MpGABR* binding pocket (Fig. 7A [↗](#)), among which merely ALA226 and THR228 interact with betulin via hydrogen bonding (Table S7). The results of the sequence alignment revealed that only THR228 was conserved across 11 species in the Aphididae family of Hemiptera (Fig. 7B [↗](#)). Furthermore, evaluation of the ability of betulin to

bind to MpGABR mutants with mutations at those four sites revealed that the ability of betulin to bind to T228R was significantly weaker than its ability to bind to the WT (**Fig. 7D and E** [↗](#), Table S8), indicating that THR228 is an essential specific site for the binding of betulin to MpGABR. Moreover, to further prove that THR228 is the specific binding site for betulin in MpGABR, the binding affinities of betulin with the WT and mutant (R122T, equivalent to THR228 in MpGABR) *Drosophila* DmGABR proteins were assessed using MST. The results showed that betulin was able to bind to DmGABR^{R122T} ($K_d = 342.4 \mu\text{M}$) but not DmGABR^{WT} (**Fig. 8B** [↗](#)). Additionally, after exposure to different concentrations of betulin, the mortality rate of DmGABR^{R122T}*Drosophila* was significantly greater than that of DmGABR^{WT}*Drosophila* (**Fig. 8E** [↗](#) and Table S9). Similarly, a previous study indicated that the R122G amino acid site substitution, generated via RNA editing, affects the sensitivity of *Drosophila* to fipronil (Es-Salah et al. 2008 [↗](#)). Together, these findings suggest that betulin binds specifically to MpGABR via THR228, acting as an inhibitor of MpGABR and causing aphid death. Studies on key amino acids that are crucial for GABR function has primarily focused on transmembrane regions. For instance, based on the mutational research and *Drosophila* GABR modeling approach, multiple key amino acids have been identified as insecticide targets in the transmembrane domain (Nakao and Banba 2021 [↗](#)). Guo et al. proposed that amino acid substitutions in transmembrane domain 2 contribute to terpenoid insensitivity during plant-insect coevolution (Guo et al. 2023 [↗](#)). However, these studies have neglected the extracellular domain. Our study signified that betulin targets the THR228 site in the extracellular domain of MpGABR, which is conserved only in the Aphididae family. Therefore, betulin is speculated to be a specific insecticidal substance evolved by plants in response to aphid infestation. Besides, further verification is needed to determine whether betulin is toxic to other insect species.

GABRs belong to the cysteine loop (Cys)-loop superfamily of neurotransmitter receptors and are essential heteropentameric ligand-gated ion channels in the central nervous system. After GABA binds to the extracellular Cys-loop of a GABR (Ashby et al. 2012 [↗](#)), the GABR is activated and opens up its central pore to allow chloride ions to pass through, thereby hyperpolarizing neurons and attenuating excitatory neurotransmission (Tremblay et al. 2016 [↗](#)). In insects, GABRs play important roles in circadian rhythms (Schellinger et al. 2022 [↗](#)), sleep (Chaturvedi et al. 2022 [↗](#)), movement (Eick et al. 2022 [↗](#)), and olfactory memory (Yamagata et al. 2021 [↗](#)). Additionally, GABR is a critical target for a variety of insecticides and ectoparasiticides. Insecticides targeting GABRs are categorized into noncompetitive antagonists (NCAs) and competitive antagonists (CAs) according to their different binding sites. NCAs block chloride channels in nerve cells by interacting with amino acid residues of the GABA-gated chloride channel, causing a conformational change in the receptor, which interferes with the normal function of the central nervous system and ultimately leads to insect death (Nakao and Banba 2021 [↗](#); Guo et al. 2023 [↗](#)). A series of first- and second-generation NCA insecticides have been successfully developed. Picrotoxinin, isolated from *Anamirta cocculus* fruit, is the oldest natural NCA and is toxic to houseflies (Tong et al. 2023 [↗](#)). Among the first-generation NCAs, polychlorocycloalkanes, including dieldrin and lindane, were commonly used as pesticides in the middle and late 20th centuries (Hainzl et al. 1998 [↗](#); Tanaka 2019 [↗](#)). As second-generation NCAs, fipronil and its derivatives are commercially available phenylpyrazole insecticides that have been widely used for agricultural pest control (Sheng et al. 2018 [↗](#); Li et al. 2021 [↗](#)). Additionally, CAs bind to orthogonal binding sites, such as the GABA recognition site in the extracellular region, competitively inhibiting the binding of GABA to its receptor, leading to toxic effects in insects. Given the lack of interference from existing insect resistance mechanisms, CAs hold promise for the development of efficient new insecticides. Novel 1, 6-dihydro-6-iminopyridazine-derived insecticides, as CAs, have insecticidal properties against common cutworms and houseflies (Liu et al. 2022 [↗](#)). Besides, nootkatone, a sesquiterpenoid, also acts as a CA to induce insect mortality (Norris et al. 2022 [↗](#)). Fluralaner, an isooxazoline ectoparasiticide, inhibits both parasites and *Aedes aegypti* by acting as a CA (Wang et al. 2022b [↗](#); Asahi et al. 2023 [↗](#)). Our results revealed that the binding site (THR228) for betulin in MpGABR was located in the extracellular neurotransmitter-gated ion-channel ligand-binding domain (Fig. S2), implying that betulin acts as a CA of MpGABR. Although the mechanism by which betulin competes with GABA for binding to

MpGABR requires further experimental validation, our work may have provided a novel target for developing insecticides. In this study, betulin, on the one hand, inhibited the expression of *MpGABR* and, on the other hand, specifically bound to MpGABR through THR228, acting as an inhibitor of MpGABR and causing aphid death (Fig. 9).

The development of bioinsecticides should not only focus on the toxic effects of active substance on target organisms, but also on their influence on the ecosystem (Haddi et al. 2020). Although our results indicate that betulin has specific toxicity to aphids, previous studies have reported that betulin and its derivatives had effects on *Plutella xylostella* L. (Huang et al. 2025), *Aedes aegypti* (de Almeida Teles et al. 2024), and *Drosophila melanogaster* (Lee and Min 2024). Therefore, further research is needed to determine whether there are other insecticidal mechanisms or off target effects of betulin. Additionally, betulin exhibits a wide range of pharmacological activities (Amiri et al. 2020), which have been used to treat various diseases, such as cancer (Lv 2023), glioblastoma (Li et al. 2022), inflammation (Szlasińska et al. 2023) and hyperlipidemia (Tang et al. 2011). Before applying betulin in the field, it is necessary to fully verify and consider whether betulin has any impact on farmers' health. Furthermore, will betulin cause residue or diffusion during the field application process? Will long-term application promote the evolution of resistance to aphids or other insects? These issues also need further experimental verification. In summary, before any field application, further research on the environmental behavior, degradation process, and safety of betulin is needed.

5. Conclusion

Betulin, a key metabolite in the aphid-resistant wild peach *P. davidiana*, possesses potent aphicidal effects on *M. persicae*. This study confirmed that betulin exhibited excellent control efficacy against *M. persicae* in both greenhouse and field experiments. RNA-seq, qRT-PCR and western blotting assays revealed that betulin significantly inhibited the expression of *MpGABR* in aphids. In addition, RNAi-mediated silencing of *MpGABR* markedly increased the sensitivity of aphids to betulin. Moreover, MST and voltage-clamp-based electrophysiological recording assays indicated that betulin was able to bind to MpGABR ($K_d = 2.24 \mu\text{M}$) and acted as an inhibitor ($\text{EC}_{50} = 20.66 \mu\text{M}$) of MpGABR. Molecular docking analyses suggested that the amino acid residue THR228, which is highly conserved across 11 species in the Aphididae family of Hemiptera, might be a critical specific binding site for betulin in MpGABR. Mutagenesis and genome editing assays revealed that betulin bound specifically to this amino acid residue in aphids but not in *Drosophila*, resulting in aphid death. Collectively, the results suggest that the aphicidal effects of betulin on aphids occur in a two-pronged manner: on the one hand, betulin inhibits *MpGABR* expression; on the other hand, it specifically binds to MpGABR via THR228 and acts as an inhibitor of MpGABR. Elucidating the insecticidal mechanism of betulin against aphids will provide a basis for the development of novel insecticides and sustainable strategies for aphid control.

Associated content

Supporting Information

Protein expression of MpGABR after RNAi (Figure S1); Betulin binding site on the domain of MpGABR (Figure S2); LC_{50} values of betulin and pymetrozine against *M. persicae* at 48 h (Table S1); Primers used for cloning, RNAi, heterologous expression, and qRT-PCR (Table S2); Oligonucleotide sequences for PCR, the sgRNA synthesis template and the fragment of donor DNA for homology-directed repair (HDR) (Table S3); Screening of genes significantly differentially expressed in *M. persicae* in response to betulin at 48 h post treatment (Table S4); Complete sequence information for the GABA_A receptor gene of *M. persicae* (Table S5); Sequences and relevant information for

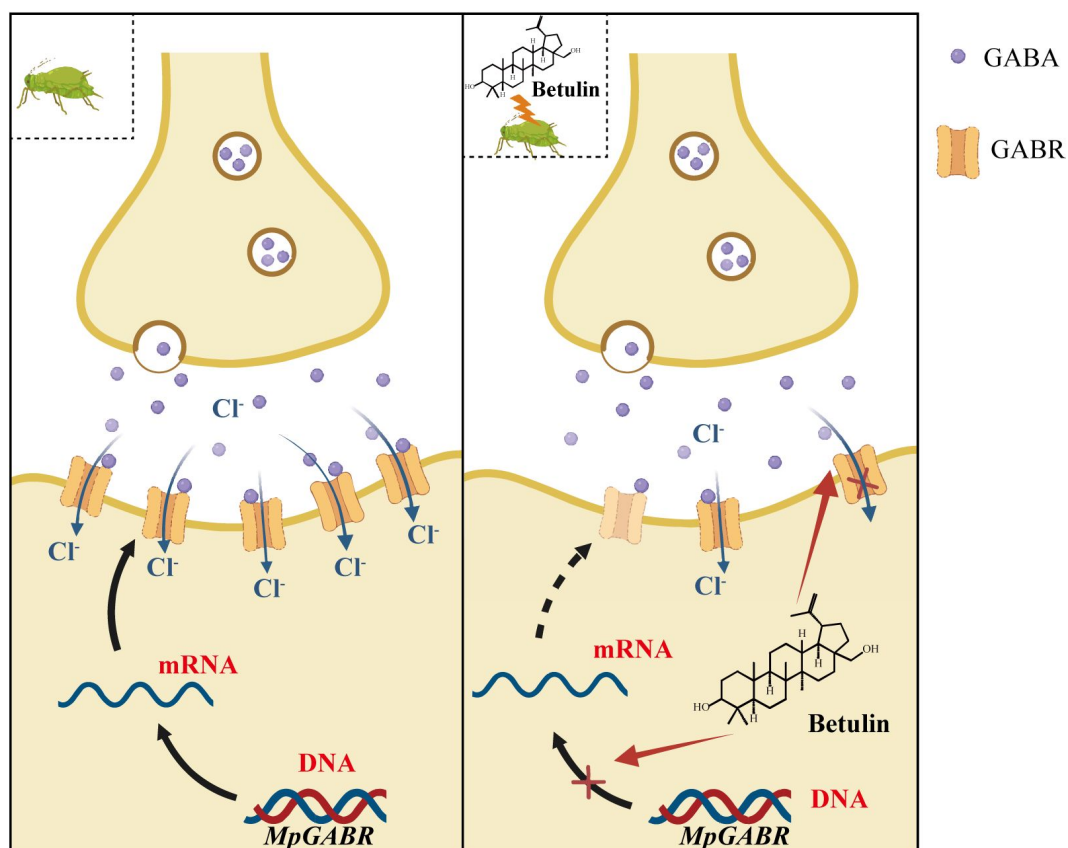


Figure 9.

Proposed model for the mechanism of action against *Myzus persicae* by targeting GABA_A receptors (GABR).

After exposure to betulin, the expression of *MpGABR* was inhibited the level of *MpGABR* protein is decreased, resulting in a decrease in the channel of chloride ion influx. Besides, betulin directly and specifically binds to the amino acid residue THR228 of *MpGABR*, thereby disabling it.

phylogenetic analysis of GABA_A receptor (Table S6); Binding energy and nonbonding interactions between betulin and GABA_A receptor (Table S7); Parameters of dose-response curves from microscale thermophoresis experiments (Table S8); LD₅₀ values of betulin and pymetrozine against *D. melanogaster* at 72 h (Table S9).

Data availability

All data generated or analyzed during this study are included in the manuscript and supporting files.

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Additional files

Supplementary Figures and Tables [↗](#)

Additional information

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Reviewer #1 (Public review):

Wang, Junxiu et al. investigated the underlying molecular mechanisms of the insecticidal activity of betulin against the peach aphid, *Myzus persicae*. There are two important findings described in this manuscript: (a) betulin inhibits the gene expression of GABA receptor in the aphid, and (b) betulin binds to the GABA receptor protein, acting as an inhibitor. The first finding is supported by RNA-Seq and RNAi, and the second one is convinced with MST and electrophysiological assays. Further investigations on the betulin binding site on the receptor protein provided a fundamental discovery that T228 is the key amino acid residue for its affinity, thereby acting as an inhibitor, backed up by site-directed mutagenesis of the heterologously-expressed receptor in *E. coli* and by CRISPR-genome editing in *Drosophila*.

Comments on revisions:

All of my review comments have been addressed, and the manuscript has been revised accordingly.

<https://doi.org/10.7554/eLife.107598.2.sa2>

Reviewer #2 (Public review):

Summary:

This important study shows that betulin from wild peach trees disrupts neural signaling in aphids by targeting a conserved site in the insect GABA receptor. The authors present a nicely integrated set of molecular, physiological, and genetic experiments to establish the compound's species-specific mode of action. While the mechanistic evidence is solid, the manuscript would benefit from a broader discussion of evolutionary conservation and potential off-target ecological effects.

Strengths:

The main strengths of the study lie in its mechanistic clarity and experimental rigor. The identification of a betulin-binding single threonine residue was supported by (1) site-directed mutagenesis and (2) functional assays. These experiments strongly support the specificity of action. Furthermore, the use of comparative analyses between aphids and fruit flies demonstrates an important effort to explore species specificity, and the integration of quantitative data further enhances the robustness of the conclusions.

Comments on revisions:

The revision satisfactorily addresses my concerns on evolutionary context, methodological clarity, and ecological risk.

<https://doi.org/10.7554/eLife.107598.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Reviewer #1 (Public review):

*Wang, Junxiu et al. investigated the underlying molecular mechanisms of the insecticidal activity of betulin against the peach aphid, *Myzus persicae*. There are two important findings described in this manuscript: (a) betulin inhibits the gene expression of GABA*

receptor in the aphid, and (b) betulin binds to the GABA receptor protein, acting as an inhibitor. The first finding is supported by RNA-Seq and RNAi, and the second one is convinced with MST and electrophysiological assays. Further investigations on the betulin binding site on the receptor protein provided a fundamental discovery that T228 is the key amino acid residue for its affinity, thereby acting as an inhibitor, backed up by site-directed mutagenesis of the heterologously-expressed receptor in *E. coli* and by CRISPR-genome editing in *Drosophila*.

Although the manuscript does have strengths in principle, the weaknesses do exist: the manuscript would benefit from more comprehensive analyses to fully support its key claims in the manuscript. In particular:

(1) The Western blotting results in Figure 5A & B appear to support the claim that betulin inhibits GABR gene expression (L26), as a decrease in target protein levels is often indicative of suppressed gene expression. The result description for Figure 5A & B is found in L312-L316, within Section 3.6 ("Responses of MpGABR to betulin"), where MST and voltage-clamp assays are also presented. It seems the observed decrease in MpGABR protein content is due to gene downregulation, rather than a direct receptor protein-betulin interaction. However, this interpretation lacks discussion or analysis in either the corresponding results section or the Discussion. In contrast, Figures 5C-F are specifically designed to illustrate protein-betulin interactions. Presenting Figure 5A & B alongside these panels might lead to confusion, as they support distinct claims (gene expression vs. protein binding/inhibition). Therefore, I recommend moving Figure 5A & B either to the end of Figure 3 or to a separate figure altogether to improve clarity and logical flow. A minor point in the Western blotting experiment is that although GAPDH was used as a reference protein, there is no explanation in the corresponding M&M section.

We thank the reviewer for the concise and accurate summary and appreciate the constructive feedback on the article's strengths and weaknesses.

(A) According to your suggestion, the original Figure 5A and B have been inserted into Figure 3, following Figure 3D. The original Figure 3E-I has been saved as a new figure, to illustrate the RNAi assay.

(b) "GAPDH was used as a reference protein" has been supplied in the M&M section, see Line 209.

(2) The description of the electrophysiological recording experiment is unclear regarding the use of GABA. I didn't realize that GABA, the true ligand of the GABA receptor, was used in this inhibition experiment until I reached the Results section (L321), which states, "In the presence of only GABA, a fast inward current was generated." Crucially, no details are provided on the experiment itself, including how GABA was applied (e.g., concentration, duration, whether GABA was treated, followed by betulin, or vice versa). This information is essential for reproducibility. Please ensure these details are thoroughly described in the corresponding M&M section.

We thank the reviewer for the valuable comments.

(a) Detailed information on how to apply GABA has been added to the corresponding M&M section (Lines 260-263): After 3 days of incubation, the oocytes were used for electrophysiological recording. GABA was dissolved in 1 × Ringer's solution to prepare 100 μM GABA solution. Subsequently, the 100 μM GABA solutions containing different concentrations of betulin (0, 5, 10, 20, 40, 80, 160, 320 μM) were used to perfuse the oocytes.

(b) Additionally, we also checked other contents of M&M section to ensure that sufficient detail has been supplied.

(3) The phylogenetic analysis, particularly concerning Figures 4 and 6B, needs significant attention for clarity and representativeness. First, your claim that MpGABR is only closely related to CAI6365831.1 (L305-L310) is inconsistent with the provided phylogenetic tree, which shows MpGABR as equally close to Metopolophium dirhodum (XP_060864885.1) and Acyrthosiphon pisum (XP_008183008.2). Therefore, singling out only Macrosiphum euphorbiae (CAI6365831.1) is not supported by the data. Second, the representation of various insect orders is insufficient. All 11 sequences in the Hemiptera category (in both Figure 4 and Figure 6B) are exclusively from the Aphididae family. This small subset cannot represent the highly diverse Order Hemiptera. Consequently, statements like "only THR228 was conserved in Hemiptera" (L338), "The results of the sequence alignment revealed that only THR228 was conserved in Hemiptera" (L430), or "THR228... is highly conserved in Hemiptera" (L486) are not adequately supported. Third, similar concerns apply to the Diptera order, which includes 10 Drosophila and 2 mosquito samples (not diverse or representative enough), and likely to other orders as well. Thereby, the Figure 6B alignment should be revised accordingly to reflect a more accurate representation or to clarify the scope of the analysis. Fourth, there's a discrepancy in the phylogenetic method used: the M&M section (L156) states that MEGA7, ClustalW, and the neighbor-joining method were used, while the Figure 4 caption mentions that MEGA X, MUSCLE, and the Maximum likelihood method were employed. This inconsistency needs to be clarified and made consistent throughout the manuscript. Fifth, I have significant concerns about the phylogenetic tree itself (Figure 4). A small glitch was observed at the Danaus plexippus node, which raises suspicion regarding potential manipulation after tree construction. More critically, the tree, especially within Coleoptera, does not appear to be clearly resolved. I am highly concerned about whether all included sequences are true GABR orthologs or if the dataset includes partial or related sequences that could distort the phylogeny. Finally, for Figure 6B, both protein (XP_) and nucleotide (XM_) sequences were mix used. I recommend using the protein sequences instead of nucleotide sequences in this figure panel, as protein sequences are more directly informative.

We thank the reviewer for the careful reading and valuable comments.

(a) Firstly, according to your comments, phylogenetic analysis has been re-performed with more represent species from each Order (Fig. 5 and Fig. 7B). The results revealed that only THR228 was conserved across 11 species in the Aphididae family of Hemiptera. Therefore, the expressions like "only THR228 was conserved in Hemiptera" have been revised to "among the four residues, only THR228 was conserved across 11 species in the Aphididae family of Hemiptera" (Line 106, Line 369, Line 477, and Lines 563-564).

(b) We have modified the description of Fig. 5 (the original Fig. 4): MpGABR (XP_022173711.1) was found to be genetically closely related to CAI6365831.1 from Macrosiphum euphorbiae, XP 008183008.2 from Acyrthosiphon pisum, and XP 060864885.1 from Metopolophium dirhodum (Fig. 5 and Table S6). See Lines 342-346.

(c) Phylogenetic analysis was performed using MEGA7 with multiple amino acid sequence alignment (ClustalW) and the neighbor-joining method. We have revised the Fig. 5 (the original Fig. 4) caption to make it accurate and consistent throughout the manuscript.

(d) We are sorry about the small glitch at the Danaus plexippus node. Actually, after the phylogenetic tree was constructed, it was imported in Adobe Illustration for coloring and classification annotation. There may have been operational errors during the process of resizing the image, resulting in the occurrence of the small glitch. Besides, the unclear

clustering of Coleoptera may be due to improper regulation of distance (pixels) of branch from nodes. Again, thanks for your careful reading. We have rebuilt the phylogenetic tree.

(e) Based on your suggestion, the sequence IDs have been unified as the protein sequence IDs (Fig. 5, Fig. 7B and Table S6)

(4) The Discussion section requires significant revision to provide a more insightful and interpretative analysis of the results. Currently, much of the section primarily restates findings rather than offering deeper discussion. For instance, L409-L419 restate the results, followed by the short sentence "Collectively, these results suggest that betulin may have insecticidal effects on aphids by inhibiting MpGABR expression". It could be further expanded to make it beneficial to elaborate on proposed mechanisms by which gene expression might be suppressed, including any potential transcription factors involved. In contrast, while L422-L442 also initially summarize results, the subsequent paragraph (L445-L472) effectively discusses the potential mechanisms of inhibitory action and how mortality is triggered, which is a good model for other parts of the section. However, all the discussion ends up with a short statement, "implying that betulin acts as a CA of MpGABR" (L472), which appears to be a leap. The inference that betulin acts as a competitive antagonist (CA) is solely based on the location of its extracellular binding site, which does not exactly overlap with the GABA binding site. It needs stronger justification or actually requires further experimental validation. The authors should consider rephrasing this statement to acknowledge the need for additional studies to definitively confirm this mechanism of action.

We appreciate the reviewer's careful reading and valuable feedback, which will certainly enhance the quality of our manuscript.

(a) Possible reasons for the effect of betulin on MpGABR expression have been discussed in our manuscript (Lines 455-466): The regulation of gene expression is sophisticated and delicate (Pope and Medzhitov 2018). The regulatory network controlling GABR expression remains unclear. In adult rats, epileptic seizures has been reported to increase the levels of brain-derived neurotrophic factor (BDNF), which in turn prompted the transcription factors CREB and ICER to reduce the gene expression of the GABR $\alpha 1$ subunit (Lund et al. 2008). In *Drosophila*, it has been demonstrated that WIDE AWAKE, which regulated the onset of sleep, interacted with the GABR and upregulated its expression level (Liu et al. 2014). In *Drosophila* brain, circular RNA circ_sxc was found to inhibit the expression of miR-87-3p in the brain through sponge adsorption, thereby regulating the expression of neurotransmitter receptor ligand proteins, including GABR, and ensuring the normal function of synaptic signal transmission in brain neurons (Li et al. 2024). However, it remains unclear how betulin reduces the expression of MpGABR, and further research is needed.

(b) In the Discussion section, we acknowledged the need for further research to ultimately confirm the mechanism by which betulin competes with GABA for binding to MpGABR (Lines 532-535): Although the mechanism by which betulin competes with GABA for binding to MpGABR requires further experimental validation, our work may have provided a novel target for developing insecticides.

(c) Besides, we have added the discussion of the sensitivity of GABA receptor to betulin in Discussion section (Lines 491-501): Studies on key amino acids that are crucial for GABR function has primarily focused on transmembrane regions. For instance, based on the mutational research and *Drosophila* GABR modeling approach, multiple key amino acids were identified as insecticide targets in the transmembrane domain (Nakao and Banba 2021). Guo et al. proposed that amino acid substitutions in the transmembrane domain 2 contribute to terpenoid insensitivity during plant-insect coevolution (Guo et al. 2023). However, these studies have neglected the extracellular domain. Our study signified that betulin targets the

THR228 site in the extracellular domain of MpGABR, which is conserved only in the Aphididae family. Therefore, betulin is speculated to be a specific insecticidal substance evolved by plants in response to aphid infestation. Besides, further verification is needed to determine whether betulin is toxic to other insect species.

(d) Furthermore, the discussion of potential ecological risks of deploying betulin as a bioinsecticide has been elaborated in our manuscript (Lines 538-553): The development of bioinsecticides should not only focus on the toxic effects of active substance on target organisms, but also on their influence on the ecosystem (Haddi et al. 2020). Although our results indicate that betulin has specific toxicity to aphids, previous studies have reported that betulin and its derivatives had effects on *Plutella xylostella* L. (Huang et al. 2025), *Aedes aegypti* (de Almeida Teles et al. 2024), and *Drosophila melanogaster* (Lee and Min 2024). Therefore, further research is needed to determine whether there are other insecticidal mechanisms or off target effects of betulin. Additionally, betulin exhibits a wide range of pharmacological activities (Amiri et al. 2020), which have been used to treat various diseases, such as cancer (Lv 2023), glioblastoma (Li et al. 2022), inflammation (Szlasa et al. 2023) and hyperlipidemia (Tang et al. 2011). Before applying betulin in the field, it is necessary to fully verify and consider whether betulin has any impact on farmers' health. Furthermore, will betulin cause residue or diffusion in the process of field application? Will long-term application promote the evolution of resistance to aphids or other insects? These issues also need further experimental verification. In summary, before any field application, further research is needed on the environmental behavior, degradation process, and safety of betulin.

Reviewer #2 (Public review):

Summary:

This important study shows that betulin from wild peach trees disrupts neural signaling in aphids by targeting a conserved site in the insect GABA receptor. The authors present a nicely integrated set of molecular, physiological, and genetic experiments to establish the compound's species-specific mode of action. While the mechanistic evidence is solid, the manuscript would benefit from a broader discussion of evolutionary conservation and

potential off-target ecological effects.

Strengths:

The main strengths of the study lie in its mechanistic clarity and experimental rigor. The identification of a betulin-binding single threonine residue was supported by (1) site-directed mutagenesis and (2) functional assays. These experiments strongly support the specificity of action. Furthermore, the use of comparative analyses between aphids and fruit flies demonstrates an important effort to explore species specificity, and the integration of quantitative data further enhances the robustness of the conclusions.

Weaknesses:

There are several important limitations that need to be addressed. The manuscript does not explore whether the observed sensitivity to betulin reflects a broadly conserved feature of GABA receptors across animal lineages or a more lineage-specific adaptation. This evolutionary context is crucial for understanding the broader significance of the findings.

In addition, while the compound's aphicidal effect is well established, the potential for off-target effects in non-target organisms - especially vertebrates - remains unaddressed, despite prior evidence that betulin interacts with mammalian GABA_A receptors. There is

little discussion on the ecological or environmental safety of exogenous betulin application, such as persistence, degradation, or exposure risks.

We sincerely thank the reviewer for the time and effort dedicated to our manuscript's detailed review and assessment. The revision suggestions were constructive, and we have provided a point-by-point response to address them.

(a) Briefly introduce the evolutionary conservation of GABA receptors has been added in the Introduction (Lines 90-98): Previous study has proposed that vertebrate and human GABR genes maintain a broad and conservative gene clustering pattern, while in invertebrates, this pattern is missing, indicating that these gene clusters formed early in vertebrate evolution and were established after diverging from invertebrates. Notably, invertebrates each possess a unique GABR gene pair, which are homologous with human GABR α and β subunits, suggesting that the existing GABR gene cluster evolved from an ancestral α - β subunit gene pair (Tsang et al. 2006). During the coevolution of plants and insects, the duplications and amino acid substitutions in GABR may be beneficial for the adaptation to insecticides and terpenoid compounds (Guo et al. 2023).

(b) We have added the discussion of the sensitivity of GABA receptor to betulin in Discussion section (Lines 491-501): Studies on key amino acids that are crucial for GABR function has primarily focused on transmembrane regions. For instance, based on the mutational research and *Drosophila* GABR modeling approach, multiple key amino acids were identified as insecticide targets in the transmembrane domain (Nakao and Banba 2021). Guo et al. proposed that amino acid substitutions in the transmembrane domain 2 contribute to terpenoid insensitivity during plant-insect coevolution (Guo et al. 2023). However, these studies have neglected the extracellular domain. Our study signified that betulin targets the THR228 site in the extracellular domain of MpGABR, which is conserved only in the Aphididae family. Therefore, betulin is speculated to be a specific insecticidal substance evolved by plants in response to aphid infestation. Besides, further verification is needed to determine whether betulin is toxic to other insect species.

(c) The discussion of potential ecological risks of deploying betulin as a bioinsecticide has been elaborated in our manuscript (Lines 538-553): The development of bioinsecticides should not only focus on the toxic effects of active substance on target organisms, but also on their influence on the ecosystem (Haddi et al. 2020). Although our results indicate that betulin has specific toxicity to aphids, previous studies have reported that betulin and its derivatives had effects on *Plutella xylostella* L. (Huang et al. 2025), *Aedes aegypti* (de Almeida Teles et al. 2024), and *Drosophila melanogaster* (Lee and Min 2024). Therefore, further research is needed to determine whether there are other insecticidal mechanisms or off target effects of betulin. Additionally, betulin exhibits a wide range of pharmacological activities (Amiri et al. 2020), which have been used to treat various diseases, such as cancer (Lv 2023), glioblastoma (Li et al. 2022), inflammation (Szlasa et al. 2023) and hyperlipidemia (Tang et al. 2011). Before applying betulin in the field, it is necessary to fully verify and consider whether betulin has any impact on farmers' health. Furthermore, will betulin cause residue or diffusion in the process of field application? Will long-term application promote the evolution of resistance to aphids or other insects? These issues also need further experimental verification. In summary, before any field application, further research is needed on the environmental behavior, degradation process, and safety of betulin.

Reviewer #1 (Recommendations for the authors):

(1) L28 Provide the full name of MST.

Thanks for your suggestion. The full name of MST, microscale thermophoresis, has been supplied.

| (2) L87 in the Order Hemiptera.

Thanks for your suggestion. Corrected.

| (3) L99 "Leaf bioassay" would be better to differentiate the greenhouse and field bioassays.

Thanks for your suggestion. Corrected.

| (4) L104 It should be 7 doses, including the "0 mg/mL" control.

Thanks for your suggestion. Corrected.

| (5) L104 Since the LC50 of pymetrozine is 1.0612 mg/mL, a wider range of doses should have been tested compared to the dose range of betulin.

Thanks for your comment.

(a) Firstly, seven doses (0, 0.0625, 0.125, 0.25, 0.5, 1, and 2 mg mL⁻¹) were set to calculate the LC50 of betulin and pymetrozine. Since the LC50 values of betulin and pymetrozine are 0.1641 and 1.0612 mg mL⁻¹, respectively, which are within the set range, indicating that the set dose range is reasonable and the LC50 values of betulin and pymetrozine are reliable.

(b) To compare the control effects of betulin and pymetrozine against *M. persicae*, LC50 of betulin (0.1641 mg mL⁻¹) and pymetrozine (1.0612 mg mL⁻¹) were used to treat *M. persicae*.

| (6) L109 Greenhouse and field bioassays.

Thanks for your suggestion. Corrected.

| (7) L112 Tween-80 and acetone in L103. Keep the order consistent throughout the manuscript.

Thanks for your suggestion. Corrected.

| (8) L122 Mortality was recorded at 1, 5, 9, and 14 days after treatment. Revise the other similar mistakes throughout the manuscript (e.g. L250, L254, L255, L256, L259, etc.).

Thanks for your suggestion. Corrected.

| (9) L126 apterous instead of wingless (keep a consistent expression).

Thanks for your suggestion. Corrected.

| (10) L138 Primer Premier?

Thanks for your comment. Corrected.

| (11) L141 Add RPS18 primers in Table S2.

Thanks for your comment. Corrected.

| (12) L155 MEGA7 vs. MEGAX (as described in the Figure 4 caption).

Thanks for your comment. Corrected.

| (13) L156 *NJ method vs. ML method (as described in the Figure 4 caption).*

Thanks for your comment. Corrected.

| (14) L157 2.7. *RNAi assay (Remove "In vitro" and re-number the following M&M sections accordingly).*

Thanks for your comment. Corrected.

| (15) L163 *Add dsGFP primers in Table S2.*

Thanks for your comment. Corrected.

| (16) L166 *apterous instead of wingless (keep a consistent expression).*

Thanks for your comment. Corrected.

| (17) L172 *Add the source of pET-B2M vector.*

pET-B2M vector was obtained from BGI (Shenzhen, China), which has been added in our manuscript (Line 194).

| (18) L195 *coding sequence instead of cDNA.*

Thanks for your comment. Corrected.

| (19) L198 *the mutations of R224A ...*

Thanks for your comment. Corrected.

| (20) L199 *TYR), or T228R ...*

Thanks for your comment. Corrected.

| (21) L211 *and 90 ng.*

Thanks for your comment. Corrected.

| (22) L213 *genomic DNA instead of gDNA, because gDNA may be confused in the context of sgRNA.*

Thanks for your suggestion. Corrected.

| (23) L253 *(Fig. 1A-B).*

Thanks for your comment. Corrected.

| (24) L268 *Explain why these 15 DEGs were selected for qRT-PCR.*

Thanks for your comment. These 15 DEGs were randomly selected and act as representative DEGs with different expression levels. The reason for selection of these 15 DEGs were added

in the manuscript (Lines 295-296).

| (25) L287 *What about GABRB? It has a TM domain.*

GABRB refers to “gamma-aminobutyric acid receptor subunit beta-like” annotated on NCBI. Theoretically, it should contain four transmembrane structural domains, while it has only one, indicating that it is incomplete.

| (26) L297 *Add dsGFP as another control group.*

Thanks for your comment. Corrected.

| (27) L299 *increased by 30.44% (Remove a comma).*

Thanks for your comment. Corrected.

| (28) L308 *XM_022318019.1 (or protein accession number with XP_).*

Thanks for your comment. Corrected.

| (29) L338 *that THR228 was conserved only in Hemiptera.*

Thanks for your comment. Since our original intention was to emphasize that THR228 is the only conserved among the four key amino acid residues, after careful consideration, we retained the expression “only THR228”.

| (30) L342 *or T228R.*

Thanks for your comment. Corrected.

| (31) L382 *Is pyridone a general name for pymetrozine?*

Thanks for your comment. Corrected.

| (32) L450 *Remove “and so on”.*

Thanks for your comment. Corrected.

| (33) *Figure 1D: Remove “Environment friendly”. Replace the plant pot image on the right side with the one sprayed with pymetrozine, like the one in Figure 1F.*

Thanks for your comment.

(a) “Environment friendly” in Figure 1D has been removed.

(b) We have attempted to modify the Figure 1D according to your suggestion. However, the modified Figure 1D is similar to Figure 1F and appears monotonous. Therefore, we have retained the original framework of Figure 1D.

| (34) *Figure 2E 111036117 and 111041856 are in different IDs (XM_). I suggest keeping GeneID in Figure 2E and Table S2, as shown in Table S4.*

Thanks for your comment. Corrected.

(35) Figure 2H: Add unit of the heatmap values. Or just add the title (e.g., expression level) on top of the bar.

Thanks for your comment. Corrected.

(36) Figure 3A: Add "aa" next to 700.

Thanks for your comment. Corrected.

(37) Figure 3E-G: Revise the tick marks on Y-axis: 0.0, 0.5, 1.0, and 1.5.

Thanks for your comment. Corrected.

(38) Figure 5C: Remove "1" and move "WT" up to the position where "1" was.

Thanks for your comment. Corrected.

(39) Figure 5D: Revise the tick marks on the Y-axis: 0.0, 0.5, 1.0, and 1.5.

Thanks for your comment. Corrected.

(40) Figure 5E: Remove the decimal. (e.g. 5 μ M, 10 μ M, 20 μ M, etc.).

Thanks for your comment. Corrected.

(41) Figure 6B: What are the numbers next to the amino acid sequences? Provide the information in the figure caption.

Thanks for your comment. The numbers next to the amino acid indicates the site of the last residue of the key amino acids, which was supplied in the figure caption.

(42) Figure 6D: Revise the tick marks on the Y-axis: 0.0, 0.5, 1.0, and 1.5. The X-axis title should be betulin (see Figure 5D). In the figure caption at the 5th row from the top, R244A should be R224A.

Thanks for your comment. Corrected.

(43) Figure 7E: R122T (not R1272T).

Thanks for your comment. Corrected.

(44) Supplementary Figure 1: It should be Figure S1. Add dsGFP in the figure caption.

Thanks for your comment. Corrected.

(45) Figure S2: What are the two pink bars and the other bars in brown or blue? Add an appropriate explanation in the figure caption.

Thanks for your comment. Corrected.

(46) Table S1: r square?

Thanks for your comment. It is "r square" and corrected.

(47) Table S2: (a) Add horizontal lines to separate qPCR, RNAi, cloning, and heterologous expression from each other (b) Replace XM_022318017.1 and XM_022318019.1 with their corresponding GeneIDs, as shown in Table S4. (c) AK340444.1 is a sequence from another aphid (*Acyrtosiphon pisum*)-Revise it. (d) In the cloning primers, place MpGABR first, followed by MpGABRAP and MpGABRB, as shown in the manuscript and Table S5. (e) Also, in the cloning primers, MpGABRB and MpGABRAP use reverse primers without stop codon, while MpGABR uses stop codon (TCA = TGA in reverse)-Revise it accordingly. Otherwise, provide the reason.

Thanks for your comment. Corrected.

(48) Table S3: (a) Add "*Drosophila melanogaster*" and the target sequence ID in the table caption. Is it KF881792.1, as shown in Table S6? (b) Align the sequences to the left side.

Thanks for your comment.

(a) The GenBank number of target sequence is KF881792.1 (*Drosophila melanogaster*). We have added this information in the Table S3 note.

(b) It has been adjusted according to your suggestion.

(49) Table S5: (a) Replace the accession numbers with GeneID, as shown in Table S4. K340444.1 is a sequence from another aphid (*Acyrtosiphon pisum*), (b) Coding sequences with stop codon are 2082, 357, and 753, respectively, while the sequences without stop codon are 2079, 354, and 750, respectively. The lengths of the deduced amino acids are 693, 118, and 250. Revise accordingly.

Thanks for your comment. Corrected.

(50) Table S6: (a) Use GenBank No for protein sequences. There is no Gene ID in this table. (b) Order (instead of Class). (c) See my comment on the phylogenetic analysis above.

Thanks for your comment. Corrected.

(51) Table S7 (a) Add unit under "Binding Energy". (b) There are two ALA226 [Alkyl] with two different distances. (c) PHE227 at the bottom should be THR228?

Thanks for your comment.

(a) The unit of "Binding Energy" was $\text{kcal}\cdot\text{mol}^{-1}$, and it was added in the table caption.

(b) Refer to Figure 6A, there were two Alkyl interaction between ALA226 and betulin. Therefore, there were two ALA226 [Alkyl] with two different distances.

(c) Similarly, there were two Pi-Alkyl interactions between PHE227 and betulin. Thus, there were two rows of PHE227 in the table.

(52) Table S9 (a) R117T should be R122T. (b) *r* square?

Thanks for your comment. a and b Corrected.

Reviewer #2 (Recommendations for the authors):

(1) Introduction

(a) It lacks a deeper biological and evolutionary framing of the GABA receptor system. As GABA receptors are highly conserved across animal taxa, the observed interaction between betulin and the aphid GABA receptor could have broader implications. This possibility is not addressed in the current version, which limits the reader's appreciation of the relevance of this mode of action.

(b) Previous reports of betulin activity in mammalian systems are not mentioned in the introduction, even though they are directly relevant to concerns about off-target toxicity. Therefore, the introduction should be revised to (i) briefly introduce the evolutionary conservation of GABA receptors, and (ii) acknowledge that betulin may affect a broader range of organisms, which sets up the need for caution in its application.

Thanks for your important suggestions.

(a) Briefly introduce the evolutionary conservation of GABA receptors has been added in the Introduction (Lines 90-98): Previous study has proposed that vertebrate and human GABR genes maintain a broad and conservative gene clustering pattern, while in invertebrates, this pattern is missing, indicating that these gene clusters formed early in vertebrate evolution and were established after diverging from invertebrates. Notably, invertebrates each possess a unique GABR gene pair, which are homologous with human GABR α and β subunits, suggesting that the existing GABR gene cluster evolved from an ancestral α - β subunit gene pair (Tsang et al. 2006). During the coevolution of plants and insects, the duplications and amino acid substitutions in GABR may be beneficial for the adaptation to insecticides and terpenoid compounds (Guo et al. 2023).

(b) The possible effects of betulin on a broader range of organisms have been acknowledged in the Introduction section (Lines 68-77): An immune stimulant, Ir-Bet, was prepared using iridium complex and betulin, which evoked ferritinophagy-enhanced ferroptosis, thereby activating anti-tumor immunity (Lv 2023). The anti-inflammatory effect of betulin has been reported in macrophages at lymphoma site in mice (Szlasa et al. 2023). Betulin has been found to improve hyperlipidemia and insulin resistance and decrease atherosclerotic plaques by inhibiting the maturation of sterol regulatory element-binding protein (Tang et al. 2011). Besides, betulin and its derivatives have been found to exhibit insecticidal activity against *Plutella xylostella* L. (Huang et al. 2025), *Aedes aegypti* (de Almeida Teles et al. 2024), and *Drosophila melanogaster* (Lee and Min 2024).

(c) At the end of the introduction, we remind that betulin should be used with caution (Lines 111-112): However, given that betulin may affect a wider range of organisms, it should be used with caution.

(2) Method

Number of biological replicates in all assays and justification of thresholds used for significance in RNAi and survival experiments are not addressed in the manuscript.

Thanks for your careful reading. We have checked Materials and Methods section and added corresponding number of biological replicates in all assays. Besides, the p-values for the corresponding significance analyses of RNAi and survival experiments have been added to our Manuscript.

(2) Discussion

(a) Consistent with the comments on the Introduction, the absence of discussion on (i) the evolutionary conservation of GABA receptor sensitivity to betulin, (ii) potential off-target effects in non-target insects and vertebrates (if so, this cannot be use for "eco-friendly

pesticide" as the authors stated in the manuscript), and (iii) ecological risks associated with the exogenous application of betulin limits both the interpretive depth and applied relevance of the study.

(b) To strengthen the Discussion, the authors should consider addressing: (i) whether the observed sensitivity reflects a conserved pharmacological vulnerability across animal taxa or a lineage-specific adaptation; (ii) the potential ecological risks of deploying betulin as a bioinsecticide, and (iii) the need for future research into the environmental fate, degradation, and safety profile of betulin prior to any field-level application.

Thank you for your valuable comments.

(a) We have added the discussion of the sensitivity of GABA receptor to betulin in Discussion section (Lines 491-501): Studies on key amino acids that are crucial for GABR function has primarily focused on transmembrane regions. For instance, based on the mutational research and *Drosophila* GABR modeling approach, multiple key amino acids were identified as insecticide targets in the transmembrane domain (Nakao and Banba 2021). Guo et al. proposed that amino acid substitutions in the transmembrane domain 2 contribute to terpenoid insensitivity during plant-insect coevolution (Guo et al. 2023). However, these studies have neglected the extracellular domain. Our study signified that betulin targets the THR228 site in the extracellular domain of MpGABR, which is conserved only in the Aphididae family. Therefore, betulin is speculated to be a specific insecticidal substance evolved by plants in response to aphid infestation. Besides, further verification is needed to determine whether betulin is toxic to other insect species.

(b) The discussion of potential ecological risks of deploying betulin as a bioinsecticide has been elaborated in our manuscript (Lines 538-551): The development of bioinsecticides should not only focus on the toxic effects of active substance on target organisms, but also on their influence on the ecosystem (Haddi et al. 2020). Although our results indicate that betulin had specific toxicity to aphids, previous studies have reported that betulin and its derivatives had effects on *Plutella xylostella* L. (Huang et al. 2025), *Aedes aegypti* (de Almeida Teles et al. 2024), and *Drosophila melanogaster* (Lee and Min 2024). Therefore, further research is needed to determine whether there are other insecticidal mechanisms or off target effects of betulin. Additionally, betulin exhibits a wide range of pharmacological activities (Amiri et al. 2020), which have been used to treat various diseases, such as cancer (Lv 2023), glioblastoma (Li et al. 2022), inflammation (Szlasa et al. 2023) and hyperlipidemia (Tang et al. 2011). Before applying betulin in the field, it is necessary to fully verify and consider whether betulin has any impact on farmers' health. Furthermore, will betulin cause residue or diffusion in the process of field application? Will long-term application promote the evolution of resistance to aphids or other insects? These issues also need further experimental verification.

(c) Additionally, at the end of the Discussion, we remind that more research is needed before any field application of betulin (Lines 551-553): In summary, before any field application, further research on the environmental behavior, degradation process, and safety of betulin is needed.

Reference

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